```
/* ================================================== */
/* 文件: game.js */
/* 弹弹珠机 - 微信小游戏入口
 * 微信开发者工具直接导入本目录即可运行(本项目根目录就是小游戏项目根)。
 * 正式上传双平台请运行 `node tools/build-minigames.js`,产出 dist/wx 与 dist/douyin
 * (BGM 压缩进分包,主包 <4MB)。
 */
require('./src/core/util.js');
require('./src/platform/platform.js');
require('./src/core/audio.js');
require('./src/game/agebadge.js');
require('./src/game/music.js');
require('./src/core/physics.js');
require('./src/game/config.js');
require('./src/game/entities.js');
require('./src/game/board.js');
require('./src/game/fx.js');
require('./src/game/ui.js');
require('./src/game/marble-game.js');
require('./src/main.js');
/* ================================================== */
/* 文件: ./src/core/util.js */
/* 弹弹珠机 - 通用工具 (命名空间: MG.util) */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  var TAU = Math.PI * 2;
  function clamp(v, a, b) { return v < a ? a : (v > b ? b : v); }
  function lerp(a, b, t) { return a + (b - a) * t; }
  function rand(a, b) { return a + Math.random() * (b - a); }
  function randInt(a, b) { return Math.floor(rand(a, b + 1)); }
  function pick(arr) { return arr[Math.floor(Math.random() * arr.length)]; }
  function dist(x1, y1, x2, y2) { var dx = x2 - x1, dy = y2 - y1; return Math.sqrt(dx * dx + dy
       * dy); }
  function fmt(n) {
    n = Math.floor(n);
    return String(n).replace(/\B(?=(\d{3})+(?!\d))/g, ',');
  }
  /* 确定性随机(用于每关布局) */
  function mulberry32(seed) {
    var t = seed >>> 0;
    return function () {
      t += 0x6D2B79F5;
      var r = Math.imul(t ^ (t >>> 15), 1 | t);
      r ^= r + Math.imul(r ^ (r >>> 7), 61 | r);
      return ((r ^ (r >>> 14)) >>> 0) / 4294967296;
    };
  }
  function easeOut(t) { t = clamp(t, 0, 1); return 1 - Math.pow(1 - t, 3); }
  function easeIn(t) { t = clamp(t, 0, 1); return t * t * t; }
```

```
  MG.util = {
    TAU: TAU,
    clamp: clamp,
    lerp: lerp,
    rand: rand,
    randInt: randInt,
    pick: pick,
    dist: dist,
    fmt: fmt,
    mulberry32: mulberry32,
    easeOut: easeOut,
    easeIn: easeIn
  };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ========================================================== */
/* 文件: ./src/platform/platform.js */
/* 弹弹珠机 - 平台适配层 (命名空间: MG.platform)
 * 一套代码适配三端:
 *   browser : 任意现代浏览器(开发/调试用)
 *   wechat  : 微信小游戏 (wx)
 *   douyin  : 抖音小游戏 (tt)
 * 对上层暴露统一 API:canvas / 尺寸 / 触摸 / 存储 / 震动 / 音效播放 / 离屏画布。
 */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  function detect() {
    if (typeof tt !== 'undefined' && tt.getSystemInfoSync) return 'douyin';
    if (typeof wx !== 'undefined' && wx.getSystemInfoSync) return 'wechat';
    return 'browser';
  }
  function metrics() {
    try {
      if (typeof tt !== 'undefined' && tt.getSystemInfoSync) {
        var t = tt.getSystemInfoSync();
        return { w: t.windowWidth || 375, h: t.windowHeight || 667, dpr: Math.min(t.pixelRatio
      || 2, 2) };
      }
      if (typeof wx !== 'undefined' && wx.getSystemInfoSync) {
        var s = wx.getSystemInfoSync();
        return { w: s.windowWidth || 375, h: s.windowHeight || 667, dpr: Math.min(s.pixelRatio
      || 2, 2) };
      }
    } catch (e) { /* fallback */ }
    var w = (typeof window !== 'undefined' && window.innerWidth) || 375;
    var h = (typeof window !== 'undefined' && window.innerHeight) || 667;
    var dpr = (typeof window !== 'undefined' && window.devicePixelRatio) || 1;
    return { w: w, h: h, dpr: Math.min(dpr, 2) };
  }
```

```
function create() {
  var name = detect();
  var isMinigame = name !== 'browser';
  var g = isMinigame ? root : (typeof window !== 'undefined' ? window : root);
  var sys = name === 'wechat' ? wx : (name === 'douyin' ? tt : null);
  var api = {
    name: name,
    isMinigame: isMinigame,
    canvas: null,
    cssW: 375, cssH: 667, dpr: 2,
    _resizeCbs: [],
    _audioPool: {}
  };
  /* ---------- 画布与尺寸 ---------- */
  if (isMinigame) {
    api.canvas = sys.createCanvas(); // 第一次调用 = 屏幕画布
  } else {
    api.canvas = document.getElementById('game');
  }
  function applySize() {
    var m = metrics();
    api.cssW = m.w; api.cssH = m.h; api.dpr = m.dpr;
    api.canvas.width = Math.round(m.w * m.dpr);
    api.canvas.height = Math.round(m.h * m.dpr);
    if (!isMinigame) {
      api.canvas.style.width = m.w + 'px';
      api.canvas.style.height = m.h + 'px';
    }
    for (var i = 0; i < api._resizeCbs.length; i++) api._resizeCbs[i](api);
  }
  api.applySize = applySize;
  api.onResize = function (cb) { api._resizeCbs.push(cb); };
  /* 画布已不含状态栏(windowHeight 已扣除),不需要额外安全区;
     胶囊坐标是屏幕坐标,换算进画布需减状态栏高度 */
  var statusBarH = 0;
  var capsuleY = 0;
  if (isMinigame) {
    try {
      if (sys.getSystemInfoSync) {
        var si = sys.getSystemInfoSync();
        if (si && si.statusBarHeight) statusBarH = si.statusBarHeight;
      }
      if (sys.getMenuButtonBoundingClientRect) {
        var mb = sys.getMenuButtonBoundingClientRect();
        if (mb && mb.height) {
          capsuleY = (mb.top + mb.bottom) / 2;   /* 画布全屏,胶囊即画布坐标 */
          api._mbRaw = 'mb=' + Math.round(mb.top) + '-' + Math.round(mb.bottom);
        }
      }
      api.iconRowAdjCss = name === 'douyin' ? 21 : 0;   /* 抖音返回的矩形比实际画的高 ~20px,校正 */
```

```
  } catch (e) { }
}
api.statusBarH = statusBarH;
api.safeTop = 0;
api.capsuleY = capsuleY;   /* 画布坐标系中的胶囊中心 */
/* 图片加载(小程序用 wx.createImage,浏览器用 Image) */
/* 抖音侧边栏复访(审核必接能力):引导用户将小游戏加入侧边栏 */
api.addToSidebar = function (cb) {
  if (name === 'douyin' && typeof tt !== 'undefined' && tt.navigateToScene) {
    try {
      tt.navigateToScene({
        scene: 'sidebar',
        success: function () { if (cb) cb(true); },
        fail: function () { if (cb) cb(false); }
      });
      return;
    } catch (e) { }
  }
  if (cb) cb(false);
};
api.createImage = function () {
  if (isMinigame && sys.createImage) return sys.createImage();
  return new Image();
};
/* ---------- 触摸输入(统一为 CSS 像素坐标) ---------- */
function normTouch(e) {
  var t = (e.touches && e.touches.length ? e.touches[0] : (e.changedTouches && e.changedTou
  ches.length ? e.changedTouches[0] : e));
  var x = t.clientX != null ? t.clientX : (t.pageX != null ? t.pageX : (t.x || 0));
  var y = t.clientY != null ? t.clientY : (t.pageY != null ? t.pageY : (t.y || 0));
  return { x: x, y: y, id: t.identifier != null ? t.identifier : 0 };
}
/* ---------- 分包加载(BGM 分包,build-minigames.js 构建时生成) ---------- */
api.loadSubpackage = function (name, cb) {
  if (!isMinigame || !sys.loadSubpackage) { if (cb) cb(); return; }
  try {
    sys.loadSubpackage({
      name: name,
      success: function () { if (cb) cb(); },
      fail: function () { if (cb) cb(); }
    });
  } catch (e) { if (cb) cb(); }
};
if (isMinigame) {
  sys.onTouchStart(function (e) { api._fire('down', normTouch(e)); });
  sys.onTouchMove(function (e) { api._fire('move', normTouch(e)); });
  sys.onTouchEnd(function (e) { api._fire('up', normTouch(e)); });
  sys.onTouchCancel && sys.onTouchCancel(function (e) { api._fire('up', normTouch(e)); });
} else {
  var cvs = api.canvas;
```

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  if (window.PointerEvent) {
    cvs.addEventListener('pointerdown', function (e) { api._fire('down', { x: e.clientX, y:
   e.clientY, id: e.pointerId }); e.preventDefault(); });
    cvs.addEventListener('pointermove', function (e) { if (e.pointerType === 'mouse' || e.i
  sPrimary) api._fire('move', { x: e.clientX, y: e.clientY, id: e.pointerId }); });
    cvs.addEventListener('pointerup', function (e) { api._fire('up', { x: e.clientX, y: e.c
  lientY, id: e.pointerId }); e.preventDefault(); });
    cvs.addEventListener('pointercancel', function (e) { api._fire('up', { x: e.clientX, y:
   e.clientY, id: e.pointerId }); });
  } else {
    cvs.addEventListener('touchstart', function (e) { api._fire('down', normTouch(e)); e.pr
  eventDefault(); }, { passive: false });
    cvs.addEventListener('touchmove', function (e) { api._fire('move', normTouch(e)); e.pre
  ventDefault(); }, { passive: false });
    cvs.addEventListener('touchend', function (e) { api._fire('up', normTouch(e)); e.preven
  tDefault(); }, { passive: false });
    cvs.addEventListener('mousedown', function (e) { api._fire('down', { x: e.clientX, y: e
  .clientY, id: 0 }); });
    cvs.addEventListener('mousemove', function (e) { api._fire('move', { x: e.clientX, y: e
  .clientY, id: 0 }); });
    cvs.addEventListener('mouseup', function (e) { api._fire('up', { x: e.clientX, y: e.cli
  entY, id: 0 }); });
  }
  document.addEventListener('touchmove', function (e) { e.preventDefault(); }, { passive: f
  alse });
}
api.volume = 0.8;   // 主音量(小游戏端音效缩放)
api._handlers = { down: [], move: [], up: [] };
api.onTouch = function (type, cb) { api._handlers[type].push(cb); };
api._fire = function (type, pt) {
  var hs = api._handlers[type];
  for (var i = 0; i < hs.length; i++) hs[i](pt);
};
/* ---------- 存储 ---------- */
api.storage = {
  get: function (key, def) {
    try {
      if (isMinigame) {
        var v = sys.getStorageSync(key);
        return (v === '' || v == null) ? def : v;
      }
      var raw = window.localStorage.getItem(key);
      return raw == null ? def : JSON.parse(raw);
    } catch (e) { return def; }
  },
  set: function (key, val) {
    try {
      if (isMinigame) sys.setStorageSync(key, val);
      else window.localStorage.setItem(key, JSON.stringify(val));
    } catch (e) { /* ignore */ }
```

```
  }
};
/* ---------- 震动 ---------- */
api.vibrate = function () {
  try {
    if (name === 'wechat' && wx.vibrateShort) wx.vibrateShort({ type: 'light' });
    else if (name === 'douyin' && tt.vibrateShort) tt.vibrateShort({});
    else if (navigator && navigator.vibrate) navigator.vibrate(20);
  } catch (e) { /* ignore */ }
};
/* ---------- 小游戏端音效播放(浏览器端用 WebAudio 合成,不走这里) ---------- */
api.playSound = function (name, vol) {
  vol = (vol || 1) * (api.volume || 0.8);
  if (!isMinigame) return false;
  try {
    var pool = api._audioPool;
    var au = pool[name];
    if (!au) {
      au = sys.createInnerAudioContext();
      au.src = 'assets/sounds/' + name + '.wav';
      au.obeyMuteSwitch = true;
      pool[name] = au;
    }
    try { au.stop(); } catch (e) { }
    au.volume = vol != null ? vol : 1;
    au.play();
    return true;
  } catch (e) { return false; }
};
/* ---------- 背景音乐(小游戏端 InnerAudio;网页端用 <audio>) ---------- */
api.createInnerAudio = function () {
  try { return sys.createInnerAudioContext(); } catch (e) { return null; }
};
/* ---------- 离屏画布(背景缓存用) ---------- */
api.createOffscreen = function (w, h) {
  var c;
  if (isMinigame) c = sys.createCanvas();
  else c = document.createElement('canvas');
  c.width = w; c.height = h;
  return c;
};
/* ---------- 分享(小游戏端) ---------- */
api.share = function (title) {
  try {
    if (name === 'wechat' && wx.shareAppMessage) {
      wx.shareAppMessage({ title: title });
      return true;
    }
    if (name === 'douyin' && tt.shareAppMessage) {
      tt.shareAppMessage({ title: title });
```

```
          return true;
        }
      } catch (e) { /* ignore */ }
      return false;
    };
    if (isMinigame) {
      try { sys.showShareMenu && sys.showShareMenu({}); } catch (e) { }
    }
    api.setPreferredFramesPerSecond = function (fps) {
      try {
        if (isMinigame && sys.setPreferredFramesPerSecond) sys.setPreferredFramesPerSecond(fps)
      ;
      } catch (e) { }
    };
    applySize();
    if (!isMinigame && typeof window !== 'undefined') {
      window.addEventListener('resize', applySize);
    }
    return api;
  }
  MG.platform = { create: create, detect: detect };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ========================================================= */
/* 文件: ./src/core/audio.js */
/* 弹弹珠机 - 音频模块 (命名空间: MG.audio)
 * 浏览器端: WebAudio 实时合成(无需任何音频文件)
 * 小游戏端: 播放 assets/sounds/*.wav (由 tools/gen-audio.js 生成)
 */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  var _platform = null;
  var _muted = false;
  var _actx = null;
  var _master = null;
  function init(platform) {
    _platform = platform;
  }
  function ensureCtx() {
    if (_muted) return null;
    if (!_actx) {
      var AC = root.AudioContext || root.webkitAudioContext;
      if (!AC) return null;
      try {
        _actx = new AC();
        _master = _actx.createGain();
        _master.gain.value = 0.5;
        _master.connect(_actx.destination);
      } catch (e) { return null; }
```

```
  }
  if (_actx.state === 'suspended') { try { _actx.resume(); } catch (e) { } }
  return _actx;
}
/* 首次用户手势时调用,解锁 iOS/安卓 WebAudio */
function unlock() { ensureCtx(); }
function tone(o) {
  var a = ensureCtx();
  if (!a) return;
  try {
    var t0 = a.currentTime + (o.delay || 0);
    var osc = a.createOscillator();
    var g = a.createGain();
    osc.type = o.type || 'sine';
    osc.frequency.setValueAtTime(Math.max(1, o.f), t0);
    if (o.f2) osc.frequency.exponentialRampToValueAtTime(Math.max(1, o.f2), t0 + o.dur);
    g.gain.setValueAtTime(0.0001, t0);
    g.gain.exponentialRampToValueAtTime(Math.max(0.001, o.vol || 0.3), t0 + 0.008);
    g.gain.exponentialRampToValueAtTime(0.0001, t0 + o.dur);
    osc.connect(g); g.connect(_master);
    osc.start(t0); osc.stop(t0 + o.dur + 0.03);
  } catch (e) { /* ignore */ }
}
/* 合成音色表(仅浏览器端使用) */
function synth(name, o) {
  o = o || {};
  var p = o.pitch || 1;
  switch (name) {
    case 'peg':
      tone({ f: 640 * p, f2: 1100 * p, dur: 0.07, type: 'triangle', vol: 0.22 });
      break;
    case 'bumper':
      tone({ f: 300, f2: 620, dur: 0.13, type: 'square', vol: 0.14 });
      tone({ f: 160, f2: 90, dur: 0.12, type: 'sine', vol: 0.3 });
      break;
    case 'mover':
      tone({ f: 240, f2: 420, dur: 0.1, type: 'triangle', vol: 0.2 });
      break;
    case 'star':
      tone({ f: 880, dur: 0.09, type: 'triangle', vol: 0.25 });
      tone({ f: 1320, dur: 0.12, type: 'triangle', vol: 0.22, delay: 0.07 });
      break;
    case 'slot':
      tone({ f: 520, f2: 260, dur: 0.2, type: 'sine', vol: 0.35 });
      tone({ f: 660, dur: 0.14, type: 'triangle', vol: 0.18, delay: 0.06 });
      break;
    case 'jackpot':
      (function () {
        var notes = [523, 659, 784, 1046, 1318];
        for (var i = 0; i < notes.length; i++) {
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          tone({ f: notes[i], dur: 0.12, type: 'triangle', vol: 0.26, delay: i * 0.085 });
        }
        tone({ f: 1568, dur: 0.3, type: 'sine', vol: 0.2, delay: 0.45 });
      })();
      break;
    case 'launch':
      tone({ f: 320, f2: 90, dur: 0.18, type: 'sawtooth', vol: 0.18 });
      tone({ f: 900, f2: 1400, dur: 0.08, type: 'sine', vol: 0.12 });
      break;
    case 'click':
      tone({ f: 660, f2: 500, dur: 0.05, type: 'square', vol: 0.16 });
      break;
    case 'over':
      tone({ f: 392, dur: 0.2, type: 'sine', vol: 0.3 });
      tone({ f: 311, dur: 0.2, type: 'sine', vol: 0.3, delay: 0.17 });
      tone({ f: 262, dur: 0.34, type: 'sine', vol: 0.3, delay: 0.34 });
      break;
    case 'record':
      (function () {
        var notes = [523, 659, 784, 1046];
        for (var i = 0; i < notes.length; i++) {
          tone({ f: notes[i], dur: 0.1, type: 'square', vol: 0.15, delay: i * 0.07 });
        }
      })();
      break;
    case 'bonus':
      tone({ f: 700, f2: 1200, dur: 0.16, type: 'triangle', vol: 0.25 });
      tone({ f: 1000, f2: 1600, dur: 0.18, type: 'triangle', vol: 0.2, delay: 0.1 });
      break;
  }
}
function play(name, o) {
  if (_muted) return;
  if (_platform && _platform.isMinigame) {
    _platform.playSound(name, (o && o.vol) || 1);
  } else {
    synth(name, o);
  }
}
function setMuted(m) { _muted = !!m; }
function toggleMuted() { _muted = !_muted; return _muted; }
function isMuted() { return _muted; }
/* 背景音乐音符(芯片音) */
function note(freq, dur, type, vol) {
  var a = ensureCtx();
  if (!a) return;
  try {
    var t0 = a.currentTime;
    var osc = a.createOscillator();
    var g = a.createGain();
```

```
      osc.type = type || 'square';
      osc.frequency.setValueAtTime(Math.max(1, freq), t0);
      g.gain.setValueAtTime(0.0001, t0);
      g.gain.exponentialRampToValueAtTime(Math.max(0.001, vol || 0.05), t0 + 0.01);
      g.gain.exponentialRampToValueAtTime(0.0001, t0 + dur);
      osc.connect(g); g.connect(_master);
      osc.start(t0); osc.stop(t0 + dur + 0.03);
    } catch (e) { }
  }
  function ready() { return !!_actx; }
  /* 主音量(0..1):控制音效合成与增益 */
  function setVolume(v) {
    if (_master) {
      try { _master.gain.value = Math.max(0, Math.min(1, v)) * 0.9; } catch (e) { }
    }
  }
  MG.audio = {
    init: init,
    unlock: unlock,
    play: play,
    note: note,
    ready: ready,
    setVolume: setVolume,
    setMuted: setMuted,
    toggleMuted: toggleMuted,
    isMuted: isMuted
  };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ================================================== */
/* 文件: ./src/game/agebadge.js */
/* 弹弹珠机 - CADPA 适龄提示标识(base64 内嵌,免文件加载) */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
MG.ageBadgeData = '…（图片数据共 372701 字符，打印版省略）';
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : t
      his));
/* ================================================== */
/* 文件: ./src/game/music.js */
/* 弹弹珠机 - 背景音乐:原曲播放器 (命名空间: MG.music)
 * 播放 assets/bgm/ 下的原曲音频文件(用户自行放置,见 assets/bgm/README.txt)。
 * 网页端:每首歌一个 <audio> 且首次解锁时全部预加载 → 切歌即时出声。
 * 小游戏端:InnerAudioContext 复用单个实例,换 src 即切。
 * 缺文件自动跳到同曲下一个格式,再不行切下一首;点击音符图标切歌。
 */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  /* 播放文件链:依次尝试(想换曲目/放原曲,编辑对应数组即可) */
```

```
var SONGS = [
  { name: '甩葱歌(MMD)', files: ['assets/bgm/01-ievan.m4a', 'assets/bgm/01-ievan.ogg', 'assets/b
    gm/01-ievan.wav'] },
  { name: '最炫民族风', files: ['assets/bgm/06-zuixuan.m4a'] },
  { name: '小苹果', files: ['assets/bgm/07-pingguo.m4a'] },
  { name: '卡农', files: ['assets/bgm/02-canon.m4a', 'assets/bgm/02-canon.wav'] },
  { name: '欢乐颂', files: ['assets/bgm/03-ode.m4a', 'assets/bgm/03-ode.wav'] },
  { name: '康康舞曲', files: ['assets/bgm/04-cancan.m4a', 'assets/bgm/04-cancan.wav'] },
  { name: '小星星', files: ['assets/bgm/05-twinkle.m4a', 'assets/bgm/05-twinkle.wav'] }
];
var platform = null;
var idx = 0;
var cache = [];       /* 网页端:每首歌一个 <audio>,预加载 */
var innerEl = null;   /* 小游戏端 InnerAudio */
var wantPlay = false;
var started = false;
var muted = false;
var vol = 0.8;
var OUT_VOL = 0.32;
function init(pl) {
  platform = pl;
  idx = (pl.storage.get('mm_song', 0) | 0) % SONGS.length;
  wantPlay = true;   /* 默认开启背景音乐(静音/切歌可控) */
}
function song() { return SONGS[idx]; }
function songCount() { return SONGS.length; }
function current() { return idx; }
function setSong(i) {
  idx = ((i % SONGS.length) + SONGS.length) % SONGS.length;
  if (wantPlay && started && !muted) play();
  return SONGS[idx].name;
}
function next() { return setSong(idx + 1); }
/* ---- 网页端:每首歌缓存一个 audio,加载期就完成文件回退 ---- */
function webEl(i) {
  if (i in cache) return cache[i];
  var a = null;
  try {
    a = new root.Audio();
    a.loop = true;
    a.preload = 'auto';
    a.volume = vol * OUT_VOL;
    var st = { fi: 0 };
    a._st = st;
    a.onerror = function () {
      st.fi++;
      if (st.fi < SONGS[i].files.length) {
        a.src = SONGS[i].files[st.fi];
        a.load();
      }
```

```
      };
      a.src = SONGS[i].files[0];
      a.load();
    } catch (e) { a = null; }
    cache[i] = a;
    return a;
  }
  /* 播第 i 首:文件链在预加载时已解析好,直接出声 */
  function webPlay(i) {
    var a = webEl(i);
    if (!a) return;
    var st = a._st;
    if (!st || st.fi >= SONGS[i].files.length) return; /* 该曲没有可用文件 */
    var p = a.play();
    if (p && p.catch) p.catch(function () { /* 自动播放被拦:等首次手势 */ });
  }
  function webPauseAll() {
    for (var i = 0; i < cache.length; i++) {
      try { if (cache[i]) cache[i].pause(); } catch (e) { }
    }
  }
  /* ---- 小游戏端 ---- */
  function innerStop() {
    try { if (innerEl) { innerEl.stop(); } } catch (e) { }
  }
  function innerPlay() {
    if (!innerEl && platform.createInnerAudio) innerEl = platform.createInnerAudio();
    if (!innerEl) return;
    innerEl.loop = true;
    innerEl.volume = vol * OUT_VOL;
    innerStop();
    innerEl.src = SONGS[idx].files[0];
    try { innerEl.play(); } catch (e) { }
  }
  function play() {
    wantPlay = true;
    if (muted) return;
    if (platform.isMinigame) innerPlay();
    else { webPauseAll(); webPlay(idx); }
  }
  function pause() {
    wantPlay = false;
    webPauseAll();
    innerStop();
  }
  /* 首次用户手势后调用:浏览器允许出声了;顺带预加载全部曲目。
     小游戏端:先把 BGM 分包(bgm1/bgm2,构建时生成)加载完再播 */
  function unlock() {
    if (started) return;
    started = true;
```

```
    if (platform.isMinigame) {
      if (platform.loadSubpackage) {
        var pkgs = ['bgm1', 'bgm2'], doneN = 0;
        var onOne = function () {
          doneN++;
          if (doneN >= pkgs.length && wantPlay && !muted) play();
        };
        for (var j = 0; j < pkgs.length; j++) platform.loadSubpackage(pkgs[j], onOne);
        return;
      }
      if (wantPlay && !muted) play();
      return;
    }
    webEl(idx);   /* 只预加载当前曲目,其余切歌时按需加载(省流量) */
    if (wantPlay && !muted) play();
  }
  function setMuted(m) {
    muted = !!m;
    if (muted) {
      /* 停声但保留 wantPlay:取消静音后音乐能接续回来 */
      webPauseAll();
      innerStop();
    } else if (wantPlay && started) {
      play();
    }
  }
  function setVolume(v) {
    vol = Math.max(0, Math.min(1, v));
    for (var i = 0; i < cache.length; i++) {
      try { if (cache[i]) cache[i].volume = vol * OUT_VOL; } catch (e) { }
    }
    try { if (innerEl) innerEl.volume = vol * OUT_VOL; } catch (e) { }
  }
  MG.music = {
    init: init,
    setVolume: setVolume,
    _debug: function () {
      var a = cache[idx];
      return {
        idx: idx, started: started, wantPlay: wantPlay, muted: muted, cached: cache.length,
        el: a ? { src: (a.src || '').split('/').pop(), paused: a.paused, volume: a.volume } : n
      ull
      };
    },
    song: song,
    songCount: songCount,
    current: current,
    setSong: setSong,
    next: next,
    play: play,
```

```
    pause: pause,
    unlock: unlock,
    setMuted: setMuted
  };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ================================================== */
/* 文件: ./src/core/physics.js */
/* 弹弹珠机 - 轻量物理引擎 (命名空间: MG.physics)
 * 只处理:圆球(动态) vs 圆(静态/移动) vs 线段(墙/隔板) vs 圆球之间的碰撞。
 * 零依赖、无 DOM,可在浏览器 / 微信小游戏 / 抖音小游戏直接运行。
 */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  function World(opts) {
    opts = opts || {};
    this.g = opts.gravity != null ? opts.gravity : 2100;
    this.maxSpeed = opts.maxSpeed != null ? opts.maxSpeed : 1500;
    this.airDrag = opts.airDrag != null ? opts.airDrag : 0.08;
    this.balls = [];
    this.circles = [];   // {x,y,r,kind,e,boost,vx,vy,disabled}
    this.segs = [];      // {x1,y1,x2,y2,half,e,kind}
    this.onEvent = null; // function({type, ...})
  }
  World.prototype.addBall = function (b) {
    b.vx = b.vx || 0; b.vy = b.vy || 0;
    b.dead = false;
    this.balls.push(b);
    return b;
  };
  World.prototype.addCircle = function (c) { this.circles.push(c); return c; };
  World.prototype.addSeg = function (s) { this.segs.push(s); return s; };
  World.prototype.step = function (dt, sub) {
    sub = sub || 1;
    var h = dt / sub;
    for (var i = 0; i < sub; i++) this._sub(h);
    // 清理 dead 球(由外部 bank 后标记)
    if (this.balls.length) {
      var alive = [];
      for (var j = 0; j < this.balls.length; j++) if (!this.balls[j].dead) alive.push(this.ball
      s[j]);
      this.balls = alive;
    }
  };
  World.prototype._sub = function (h) {
    var balls = this.balls, i, b;
    var damp = 1 - this.airDrag * h;
    for (i = 0; i < balls.length; i++) {
      b = balls[i];
```

```
    if (b.dead) continue;
    b.vy += this.g * h;
    b.vx *= damp; b.vy *= damp;
    if (b.nudgeCd > 0) b.nudgeCd -= h;
    var sp = Math.sqrt(b.vx * b.vx + b.vy * b.vy);
    if (sp > this.maxSpeed) { var k = this.maxSpeed / sp; b.vx *= k; b.vy *= k; }
    b.x += b.vx * h; b.y += b.vy * h;
  }
  // 球 vs 圆(钉/弹射器/巡逻钉)
  for (i = 0; i < balls.length; i++) {
    b = balls[i];
    if (b.dead) continue;
    var cs = this.circles;
    for (var j = 0; j < cs.length; j++) {
      var c = cs[j];
      if (c.disabled) continue;
      var dx = b.x - c.x, dy = b.y - c.y;
      var R = b.r + c.r;
      var d2 = dx * dx + dy * dy;
      if (d2 >= R * R) continue;
      var d = Math.sqrt(d2) || 0.0001;
      var nx = dx / d, ny = dy / d;
      b.x = c.x + nx * R; b.y = c.y + ny * R;
      var mvx = c.vx || 0, mvy = c.vy || 0;
      var rvx = b.vx - mvx, rvy = b.vy - mvy;
      var vn = rvx * nx + rvy * ny;
      if (vn < 0) {
        var e = c.e != null ? c.e : (c.kind === 'bumper' ? 0.35 : 0.5);
        /* 微小撞击不反弹:防止球在钉顶 perpetually 微颤卡住 */
        if (-vn < 40) e = 0;
        rvx -= (1 + e) * vn * nx;
        rvy -= (1 + e) * vn * ny;
        b.vx = rvx + mvx; b.vy = rvy + mvy;
        if (c.kind === 'bumper') {
          var spN = b.vx * nx + b.vy * ny;
          var boost = c.boost || 0;
          if (spN < boost) { b.vx += (boost - spN) * nx; b.vy += (boost - spN) * ny; }
          /* 微触不发声:球停在柱顶时重力每子步压入 2.8px/s,不设阈值事件会每秒刷 720 次 */
          if (-vn > 60) this._emit({ type: 'bumper', x: c.x, y: c.y, ball: b, circle: c, spee
  d: -vn });
        } else if (c.kind === 'peg') {
          if (-vn > 60) this._emit({ type: 'peg', x: c.x, y: c.y, ball: b, circle: c, speed:
  -vn });
          /* 钉顶平衡破坏:球 nearly 静止在钉顶附近时,沿肩部向下滑出(真机圆钉不会卡球)。
             必须带冷却:step(dt,12) 每帧 12 个子步,若球被钉+墙夹住反复轻触,
             不设冷却会每子步 +130/+30 速度 → 一帧内凭空加速到数千(弹射泵) */
          if (-vn < 120 && -vn > 25 && dy < 0 && !(b.nudgeCd > 0)) {
            var side = c.slideDir || (dx >= 0 ? 1 : -1);
            b.vx += side * 130;
            b.vy += 30;
```

```
            b.nudgeCd = 0.4;
          }
        } else if (c.kind === 'mover') {
          this._emit({ type: 'mover', x: c.x, y: c.y, ball: b, circle: c, speed: -vn });
        }
      }
    }
  }
  // 球 vs 球
  for (i = 0; i < balls.length; i++) {
    var b1 = balls[i];
    if (b1.dead) continue;
    for (var k = i + 1; k < balls.length; k++) {
      var b2 = balls[k];
      if (b2.dead) continue;
      var ddx = b2.x - b1.x, ddy = b2.y - b1.y;
      var RR = b1.r + b2.r;
      var dd2 = ddx * ddx + ddy * ddy;
      if (dd2 >= RR * RR || dd2 === 0) continue;
      var dd = Math.sqrt(dd2);
      var nnx = ddx / dd, nny = ddy / dd;
      var overlap = RR - dd;
      b1.x -= nnx * overlap / 2; b1.y -= nny * overlap / 2;
      b2.x += nnx * overlap / 2; b2.y += nny * overlap / 2;
      var rv = (b2.vx - b1.vx) * nnx + (b2.vy - b1.vy) * nny;
      if (rv < 0) {
        var imp = -(1 + 0.68) * rv / 2;
        b1.vx -= imp * nnx; b1.vy -= imp * nny;
        b2.vx += imp * nnx; b2.vy += imp * nny;
        this._emit({ type: 'ballball', x: b1.x + nnx * b1.r, y: b1.y + nny * b1.r });
      }
    }
  }
  // 球 vs 线段(墙 / 隔板)
  for (i = 0; i < balls.length; i++) {
    b = balls[i];
    if (b.dead) continue;
    var sg = this.segs;
    for (var m = 0; m < sg.length; m++) {
      var s = sg[m];
      var sx = s.x2 - s.x1, sy = s.y2 - s.y1;
      var len2 = sx * sx + sy * sy;
      var t = len2 > 0 ? ((b.x - s.x1) * sx + (b.y - s.y1) * sy) / len2 : 0;
      t = t < 0 ? 0 : (t > 1 ? 1 : t);
      var px = s.x1 + sx * t, py = s.y1 + sy * t;
      var ex = b.x - px, ey = b.y - py;
      var RS = b.r + s.half;
      var e2 = ex * ex + ey * ey;
      if (e2 >= RS * RS) continue;
      var ed = Math.sqrt(e2);
```

```
      var qnx, qny;
      if (ed > 0.0001) { qnx = ex / ed; qny = ey / ed; }
      else { var sl = Math.sqrt(len2) || 1; qnx = -sy / sl; qny = sx / sl; }
      b.x = px + qnx * RS; b.y = py + qny * RS;
      var svn = b.vx * qnx + b.vy * qny;
      if (svn < 0) {
        var se = s.e != null ? s.e : 0.4;
        b.vx -= (1 + se) * svn * qnx;
        b.vy -= (1 + se) * svn * qny;
        // 切向摩擦
        var tvx = b.vx - (b.vx * qnx + b.vy * qny) * qnx;
        var tvy = b.vy - (b.vx * qnx + b.vy * qny) * qny;
        var fr = s.friction != null ? s.friction : 0.995;
        b.vx -= tvx * (1 - fr); b.vy -= tvy * (1 - fr);
        if (s.kind === 'wall' && -svn > 380) {
          this._emit({ type: 'wall', x: px, y: py, ball: b, speed: -svn });
        }
      }
      /* 端点圆帽防栖:低速球栖在段端头圆帽上时,沿帽面切向推出去,
         绕过端头后顺杆身滑落——场地侧的滑回场地,弹道侧的滑回弹道 */
      if ((t <= 0 || t >= 1) && (b.vx * b.vx + b.vy * b.vy) < 25600 && !(b.nudgeCd > 0)) {
        var sln = Math.sqrt(len2) || 1;
        var udx = sx / sln, udy = sy / sln;
        var capDx = b.x - px, capDy = b.y - py;          /* 相对端帽中心 */
        var side = (Math.abs(capDx) > 0.5 || Math.abs(capDy) > 0.5)
          ? ((capDx * udy - capDy * udx) > 0 ? 1 : -1)   /* 朝球已偏的一侧绕出 */
          : -1;                                          /* 完全居中:默认向场地一侧 */
        b.vx += -qny * side * 120;
        b.vy += qnx * side * 120;
        b.nudgeCd = 0.35;
      }
    }
  }
  /* 结算后统一限速:夹缝里多次反射叠加的速度不允许带出子步 */
  for (i = 0; i < balls.length; i++) {
    b = balls[i];
    if (b.dead) continue;
    var sp2 = Math.sqrt(b.vx * b.vx + b.vy * b.vy);
    if (sp2 > this.maxSpeed) { var k2 = this.maxSpeed / sp2; b.vx *= k2; b.vy *= k2; }
  }
};
World.prototype._emit = function (e) {
  if (this.onEvent) this.onEvent(e);
};
/* 轨迹预演:用同一套碰撞逻辑跑一个临时球,返回采样点(用于瞄准虚线) */
function simulatePath(x, y, vx, vy, circles, segs, opts) {
  opts = opts || {};
  var w = new World({ gravity: opts.g != null ? opts.g : 2100, maxSpeed: opts.maxSpeed != nul
    l ? opts.maxSpeed : 1500, airDrag: opts.airDrag != null ? opts.airDrag : 0.08 });
  w.circles = circles || [];
```

```
    w.segs = segs || [];
    var b = w.addBall({ x: x, y: y, vx: vx, vy: vy, r: opts.r || 15 });
    var dt = opts.dt || (1 / 120);
    var maxSteps = opts.maxSteps || 260;
    var pts = [];
    for (var i = 0; i < maxSteps; i++) {
      w.step(dt, 1);
      if (i % 3 === 0) pts.push({ x: b.x, y: b.y });
      if (b.y > (opts.stopY || 1e9)) break;
      var sp = Math.sqrt(b.vx * b.vx + b.vy * b.vy);
      if (i > 25 && sp < 55) break;
    }
    return pts;
  }
  MG.physics = {
    World: World,
    simulatePath: simulatePath
  };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ========================================================== */
/* 文件: ./src/game/config.js */
/* 弹弹珠机 - 全局配置与调参 (命名空间: MG.config)
 * 复刻「玻璃珠自动贩卖游艺机」:
 *   - 右下角拉杆把弹珠往上弹,沿右侧弹道冲过顶部导弧拐入斜面场地
 *   - 场地斜面约 45°(低重力慢滚),弹珠穿过 LED 钉阵滚落
 *   - 右下角自选投入弹珠数(1~5 颗)= 选倍率 ×2~×10;
 *     投入越多亮灯的洞越少;从亮灯洞滑落赢回 投入×倍率 颗弹珠
 *   - 洞口没有分数,只负责亮灯
 */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  var CFG = {
    /* 逻辑分辨率:宽度固定 750,高度按屏幕比例自适应 */
    DESIGN_W: 750,
    H_MIN: 1200,
    H_MAX: 1560,
    /* 物理:斜面滚珠(低重力慢滚),弹道爬升需要较高初速 */
    gravity: 2000,        // 斜面下滑分量(真实 45° 倾斜比例)
    maxSpeed: 3400,         // 全局限速
    airDrag: 0.1,
    ballR: 15,
    pegR: 9,
    /* 布局 */
    layout: {
      backH: 323,           // 顶部标题面板高度(底缘 = 顶轨上缘 323,与轨相接,不浮空)
      panelH: 250,          // 底部操作台
      fieldX0: 21,          // 场地左缘(左墙轨中心,轨带 14..28 与白框相接,沿边界)
      ceilingY: 330,        // 顶部导轨(轨带上缘与标题面板底缘齐平)
```

```
    cornerR: 64,          // 左右导流圆弧角(缩小:两侧导轨大部分竖直,只留小圆角)
    laneW: 80,            // 右侧弹道宽(内墙 face 到右墙 face 40px,球径 30 顺滑通过)
    laneTop: 445,         // 弹道内墙顶端,弹珠越过它进入场内
    pegTop: 430,
    pegBottomPad: 160,
    bumpers: { r: 17, boost: 0, fx: [0.22, 0.7, 0.45], fy: [0.3, 0.42, 0.68] }
  },
  /* 底部洞口:13 个,没有分数,只会亮灯 */
  pockets: {
    count: 13,
    capOff: 46
  },
  /* 倍率档位:右下角自选投入弹珠数 → 倍率 / 亮灯数 */
  tiers: [
    { cost: 1, mult: 2, lit: 5 },
    { cost: 2, mult: 3, lit: 4 },
    { cost: 3, mult: 4, lit: 3 },
    { cost: 4, mult: 5, lit: 2 },
    { cost: 5, mult: 6, lit: 1 }
  ],
  winBase: 0,             // 中奖直接赢 投入×倍率 颗弹珠(无额外基础分)
  giftCardMult: 6,        // ×6 及以上中奖送礼品卡
  startBank: 20,          // 每局初始弹珠库
  /* 发射(右下角,竖直向上) */
  launcher: {
    minSpeed: 500,        // 0% = 只弹起一点点,滚回退还
    maxSpeed: 3400,       // 100% = 掠过顶轨直到最左
    crestPower: 0.4,      // 蓄力 40% = 恰好飞出发射区(前段曲线终点)
    crestSpeed: 1900,     // 40% 时的临界速度(实测标定);后 60% 线性增大到 maxSpeed
    cooldown: 0.3,
    tapPower: 0.05,       // 快速点按 = 5% 轻弹(必须蓄力才有速度)
    chargeSpeed: 0.8,
    tapJudge: 0.16,
    refundSlow: 60,       // 弹道内低速视为滚回
    refundTime: 0.4
  },
  /* 计分 */
  scoring: {
    bumper: 0,            // 大弹柱不再给分,纯物理乐趣
    bonusScore: 0         // 已由弹珠库经济取代
  },
  colors: {},
  themeName: 'min',
  /* 双主题配色:极简(默认)/ 复古街机(可选,setTheme 原地覆写 colors) */
  palettes: {
    min: {
      /* 机台(暖白极简) */
      cabYellow: '#f6f4ef', cabYellowHi: '#fbfaf7', cabYellowLo: '#e7e4da',
      frame: '#ffffff', panelText: '#3a3a3a',
      fieldTop: '#f3f1ea', fieldBot: '#eeebe2',
```

```
    star: 'rgba(255,255,255,0.8)', glassShine: 'rgba(255,255,255,0.30)',
    ledRed: '#e8553f', ledGreen: '#58b26a', ledDim: 'rgba(0,0,0,0.10)',
    segBg: '#1d1d1f', segFrame: 'rgba(255,255,255,0.06)', segRed: '#ff5340',
    pegBody: '#ffffff', pegRim: 'rgba(0,0,0,0.10)', pegFlash: '#ffffff',
    bumperRing: 'rgba(0,0,0,0.12)',
    rail: '#2f2d29', railHi: '#3a3833', holeDark: '#141311', holeLit: '#ff4a38',
    marbleTints: ['#7ec8ff', '#ff9ecb', '#b5e8a0', '#ffd966', '#c9a8ff', '#ff8a5c'],
    text: '#3a3a3a', cream: '#ffffff', dim: '#9a9a9a', gold: '#ffcf5e', red: '#e8553f',
    /* 主题差异件 */
    laneArrow: 'rgba(0,0,0,0.14)', laneSlot: 'rgba(0,0,0,0.045)',
    laneEdge: '#dcd8cc', capDot: '#dcd8cc',
    trackMain: '#e0dcd1', trackHi: 'rgba(255,255,255,0.85)',
    dividerLine: 'rgba(255,255,255,0.28)', capFill: '#ffffff', capStroke: 'rgba(0,0,0,0.12)
',
    holeStroke: 'rgba(255,255,255,0.10)',
    ledLabel: 'rgba(255,255,255,0.4)', finePrint: 'rgba(58,58,58,0.55)',
    barTrack: 'rgba(0,0,0,0.07)',
    goBg: '#3a3a3a', goPress: '#2e2e2e', goRing: null,
    selBg: '#3a3a3a', selSub: 'rgba(255,255,255,0.7)',
    pillBg: '#ffffff', pillLine: 'rgba(0,0,0,0.08)', pillText: '#3a3a3a',
    pillDis: 'rgba(0,0,0,0.04)', pillDisText: 'rgba(0,0,0,0.28)',
    pow1: '#c9c5bc', pow2: '#3a3a3a', pow3: '#e8553f',
    pulseColor: '#ffffff', pulseRing: 'rgba(58,58,58,'
  },
  arcade: {
    /* 机台(金黄街机) */
    cabYellow: '#f9b234', cabYellowHi: '#ffcf5e', cabYellowLo: '#e0961f',
    frame: '#f8f4ea', panelText: '#6b4409',
    fieldTop: '#f8e3ad', fieldBot: '#eccf85',
    star: 'rgba(255,255,255,0.8)', glassShine: 'rgba(255,255,255,0.30)',
    ledRed: '#ff4a3a', ledGreen: '#3ce06a', ledDim: 'rgba(120,80,30,0.35)',
    segBg: '#1c0604', segFrame: '#3a1508', segRed: '#ff3222',
    pegBody: 'rgba(255,255,255,0.85)', pegRim: 'rgba(166,120,52,0.75)', pegFlash: '#ffffff'
,
    bumperRing: 'rgba(166,120,52,0.9)',
    rail: '#8a5a22', railHi: '#a5712e', holeDark: '#31200c', holeLit: '#ff3b2e',
    marbleTints: ['#7ec8ff', '#ff9ecb', '#b5e8a0', '#ffd966', '#c9a8ff', '#ff8a5c'],
    text: '#6b4409', cream: '#fff8ea', dim: '#a8813f', gold: '#ffcf5e', red: '#e8352e',
    /* 主题差异件 */
    laneArrow: 'rgba(255,209,102,0.5)', laneSlot: 'rgba(120,80,30,0.14)',
    laneEdge: '#e0961f', capDot: '#ffcf5e',
    trackMain: '#e0961f', trackHi: '#ffcf5e',
    dividerLine: 'rgba(90,55,15,0.55)', capFill: '#fdf9f2', capStroke: '#c9973f',
    holeStroke: 'rgba(255,255,255,0.28)',
    ledLabel: 'rgba(255,120,100,0.75)', finePrint: 'rgba(107,68,9,0.8)',
    barTrack: 'rgba(90,55,15,0.25)',
    goBg: '#e0392e', goPress: '#c62f2f', goRing: '#ffffff',
    selBg: '#e8352e', selSub: '#ffd9c9',
    pillBg: '#fff8ea', pillLine: '#e8b45a', pillText: '#6b4409',
    pillDis: 'rgba(120,90,50,0.35)', pillDisText: 'rgba(107,68,9,0.5)',
```

```
        pow1: '#7ec850', pow2: '#ffb84d', pow3: '#ff5c47',
        pulseColor: '#ffd166', pulseRing: 'rgba(255,255,255,'
      }
    },
    setTheme: function (name) {
      var p = this.palettes[name] || this.palettes.min;
      var c = this.colors;
      for (var k in p) c[k] = p[k];
      this.themeName = (name === 'arcade') ? 'arcade' : 'min';
    }
  };
  CFG.setTheme('min');
  MG.config = CFG;
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ================================================== */
/* 文件: ./src/game/entities.js */
/* 弹弹珠机 - 实体 (命名空间: MG.entities) */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  var CFG = MG.config;
  /* 一颗弹珠(动态球体,物理世界里由 World 驱动) */
  function Marble(x, y, vx, vy, skin) {
    this.x = x; this.y = y;
    this.vx = vx; this.vy = vy;
    this.r = CFG.ballR;
    this.skin = skin || 0;
    this.points = 0;      // 这颗弹珠本局攒到的分(落袋时乘槽位倍数)
    this.dead = false;
    this.age = 0;
    this.slowT = 0;       // 低速计时(卡住检测)
    this.nudges = 0;      // 被顶开次数
    this.trail = [];
    this.trailT = 0;
    this.wallSndT = 0;
  }
  MG.entities = { Marble: Marble };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ================================================== */
/* 文件: ./src/game/board.js */
/* 弹弹珠机 - 机器布局生成 (命名空间: MG.board)
 * 复刻玻璃珠游艺机布局:
 *   - 右下角弹道:拉杆把弹珠竖直向上弹,冲过顶部导弧拐入场内
 *   - 斜面场地:暖黄玻璃,四角圆弧(右上为弹道导弧,左上为拐弯弧)
 *   - 钉阵(部分带红/绿 LED)+ 3 个透明大弹柱
 *   - 底部 13 个洞口:没有分数,只负责亮灯(亮灯由游戏层按投入档位控制)
 */
(function (root) {
```

```
'use strict';
var MG = root.MG || (root.MG = {});
var CFG = MG.config;
var U = MG.util;
function build(round, W, H, rng) {
  var L = CFG.layout;
  var backH = L.backH;
  var panelH = L.panelH;
  var fieldX0 = L.fieldX0;
  /* ---- 右侧弹道(右下角向上,单珠宽) ---- */
  var laneX0 = W - L.laneW;
  var laneX1 = W - 15;
  var laneCx = (laneX0 + 8 + (W - 28)) / 2; /* 弹道内 face(laneX0+8) 与右墙内 face(W-28) 的正中 */
  /* ---- 场地几何 ---- */
  var floorY = H - panelH - 14;
  var fieldRight = laneX0 - 10;
  /* ---- 右侧弹道(右下角向上,单珠宽;底部与场地底线齐平) ---- */
  var laneFloorY = floorY;
  var spawn = { x: laneCx, y: laneFloorY - 42 };
  /* ---- 底部洞口(13 个,无分数):整体右移避开左侧导轨边框 ---- */
  var n = CFG.pockets.count;
  var innerX0 = fieldX0 + 7;
  var pw = (fieldRight - innerX0) / n;
  var holes = [];
  for (var i = 0; i < n; i++) {
    holes.push({
      x0: innerX0 + i * pw, x1: innerX0 + (i + 1) * pw,
      cx: innerX0 + (i + 0.5) * pw,
      lit: false, flash: 0
    });
  }
  var capY = floorY - CFG.pockets.capOff;
  /* 洞间隔针:低弹性 + 近零摩擦(摩擦按子步结算,0.92 会把球瞬间粘死) */
  var dividers = [];
  for (var d = 1; d < n; d++) {
    var dx = innerX0 + d * pw;
    dividers.push({ x1: dx, y1: capY - 40, x2: dx, y2: floorY, half: 3, e: 0.08, friction: 0.
    995, kind: 'divider' });
  }
  /* ---- 钉阵 ---- */
  var pegTop = L.pegTop;
  var pegBottom = floorY - L.pegBottomPad;
  var rows = U.clamp(Math.floor((pegBottom - pegTop) / 92), 4, 7);   /* 上限 7 排:长屏不再堆满钉子 */
  var rowGap = (pegBottom - pegTop) / (rows - 1);
  var fieldW = fieldRight - fieldX0;
  var spacing = fieldW / 8;
  var pegs = [];
  for (var r = 0; r < rows; r++) {
    var count = (r % 2 === 0) ? 8 : 7;
    var off = (r % 2 === 0) ? spacing / 2 : spacing;
```

```
    var y = pegTop + r * rowGap + (rng() - 0.5) * 10;
    for (var c = 0; c < count; c++) {
      var x = fieldX0 + off + c * spacing + (rng() - 0.5) * 10;
      /* 两墙各留 36px 通球通道(缝 < 球径必然卡死);趣味由墙边斜挡板提供 */
      x = U.clamp(x, fieldX0 + 52, fieldRight - 43);
      var led = rng();
      pegs.push({
        x: x, y: y, r: CFG.pegR, flash: 0,
        led: led < 0.28 ? 'red' : (led < 0.52 ? 'green' : null),
        /* 卡珠解救的滑落方向:靠近弹道的钉往左滑入场内,靠左墙的往右,其余随机 */
        slideDir: (x > fieldRight - 150) ? -1 : ((x < fieldX0 + 150) ? 1 : (rng() < 0.5 ? -1
    : 1))
      });
    }
  }
  /* ---- 大弹柱 ---- */
  var bumpers = [];
  var BF = L.bumpers;
  for (var bi = 0; bi < BF.fx.length; bi++) {
    bumpers.push({
      x: fieldX0 + 34 + BF.fx[bi] * (fieldW - 68),
      y: pegTop + BF.fy[bi] * (pegBottom - pegTop),
      r: BF.r, kind: 'bumper', boost: BF.boost, pulse: 0
    });
  }
  /* 清理与弹柱冲突的钉子 */
  var cleared = [];
  for (var pi = 0; pi < pegs.length; pi++) {
    var peg = pegs[pi];
    var bad = false;
    for (var q = 0; q < bumpers.length; q++) {
      /* 大弹柱与小钉之间必须留出 > 一个球直径的缝:球(直径 30)既能通过,也不会被夹死 */
      if (U.dist(peg.x, peg.y, bumpers[q].x, bumpers[q].y) < bumpers[q].r + peg.r + CFG.ballR
     * 2 + 6) { bad = true; break; }
    }
    if (!bad) cleared.push(peg);
  }
  pegs = cleared;
  /* ---- 弧段工具 ---- */
  function arcSegs(cx, cy, radius, a0, a1, steps) {
    var out = [];
    for (var s = 0; s < steps; s++) {
      var b0 = a0 + (a1 - a0) * (s / steps);
      var b1 = a0 + (a1 - a0) * ((s + 1) / steps);
      out.push({
        x1: cx + Math.cos(b0) * radius, y1: cy + Math.sin(b0) * radius,
        x2: cx + Math.cos(b1) * radius, y2: cy + Math.sin(b1) * radius,
        half: 8, e: 0.05, kind: 'rail'   /* 近无弹跳:滑行且不与侧墙形成弹射泵 */
      });
    }
```

```
  return out;
}
/* ---- 墙体 ---- */
var R = L.cornerR;
var ceilingY = L.ceilingY;
/* 两只对称的导流圆弧角:左上(横扫到边顺弧滚下)/ 右上(弹道冲顶顺弧拐入,
   右弧延长到 100°:出口略上仰,高速球靠惯性压弧贴行、出弧后贴顶滑翔,全真实碰撞) */
var cornerL = { cx: fieldX0 + R, cy: ceilingY + R, r: R };   // 弧与左墙轨(x=fieldX0)相切
var cornerR2 = { cx: W - 21 - R, cy: ceilingY + R, r: R };   // 弧与右墙轨(x=W-21)相切
/* 碰撞面与绘制轨对齐:half=7 → 胶囊面 = 画线中心±7 = 14px 轨的内缘,球缘贴轨走。
   两轨中心距机缘 21px(轨带与白框相接,沿边界走) */
var rightWallCx = W - 21;   /* 右墙轨中心 = 右上导弧端点,竖直向下 */
var walls = [
  { x1: fieldX0, y1: ceilingY, x2: fieldX0, y2: floorY, half: 7, e: 0.45, kind: 'wall' },
    // 左墙
  { x1: rightWallCx, y1: ceilingY, x2: rightWallCx, y2: laneFloorY, half: 7, e: 0.4, kind:
  'wall' }, // 右墙(机身)
  { x1: cornerL.cx, y1: ceilingY, x2: cornerR2.cx, y2: ceilingY, half: 7, e: 0.05, kind: 'w
  all' }, // 顶部导轨(贴轨滑行)
  { x1: laneX0, y1: L.laneTop, x2: laneX0, y2: laneFloorY, half: 8, e: 0.05, kind: 'wall' }
  ,  // 弹道内墙(近无弹跳)
  { x1: fieldX0, y1: floorY, x2: W - 15, y2: floorY, half: 8, e: 0.2, kind: 'wall' }
    // 场地地面(含弹道底)
];
var arcTL = arcSegs(cornerL.cx, cornerL.cy, R, -Math.PI / 2, -Math.PI, 6); // 左弧角:顶轨端→左墙端
arcTL.forEach(function (sg) { sg.e = 0.45; sg.kind = 'wall'; }); /* 左上角:普通弹性反射,把横扫的球挡
回场内 */
var arcTR = arcSegs(cornerR2.cx, cornerR2.cy, R, 0, -Math.PI / 2, 6); // 右弧角:右墙端→顶轨端(小圆
角,无上
  仰唇)
for (var a1 = 0; a1 < arcTL.length; a1++) walls.push(arcTL[a1]);
for (var a2 = 0; a2 < arcTR.length; a2++) walls.push(arcTR[a2]);
/* 墙边斜挡板:贴墙直落的球被斜坡拨回场内(球能从下方通道继续走,不卡) ---- */
var deflectors = [];
var deflectYs = [pegTop + 40, pegTop + 220, pegTop + 400];
for (var dy = 0; dy < deflectYs.length; dy++) {
  var yy = deflectYs[dy];
  if (yy > pegBottom - 60) break;
  /* 左墙:从墙面向右下 34px,坡度让球向内滑 */
  deflectors.push({ x1: fieldX0 + 20, y1: yy, x2: fieldX0 + 54, y2: yy + 26, half: 5, e: 0.
  15, friction: 0.995, kind: 'deflector' });
  /* 右墙(弹道内墙)镜像 */
  deflectors.push({ x1: laneX0 - 9, y1: yy, x2: laneX0 - 43, y2: yy + 26, half: 5, e: 0.15,
   friction: 0.995, kind: 'deflector' });
}
for (var dc = 0; dc < deflectors.length; dc++) walls.push(deflectors[dc]);
/* 挡板与钉子之间不许有夹缝:距离 < 球径+6 的钉子移除(球进不去就不存在楔死) */
var kept2 = [];
for (var pq = 0; pq < pegs.length; pq++) {
  var pg = pegs[pq], near = false;
  for (var dq = 0; dq < deflectors.length; dq++) {
    var dg = deflectors[dq];
```

```
        var vx2 = dg.x2 - dg.x1, vy2 = dg.y2 - dg.y1;
        var tt2 = ((pg.x - dg.x1) * vx2 + (pg.y - dg.y1) * vy2) / (vx2 * vx2 + vy2 * vy2);
        tt2 = tt2 < 0 ? 0 : (tt2 > 1 ? 1 : tt2);
        var qx = dg.x1 + vx2 * tt2, qy = dg.y1 + vy2 * tt2;
        if (U.dist(pg.x, pg.y, qx, qy) < pg.r + dg.half + CFG.ballR * 2 + 6) { near = true; bre
      ak; }
      }
      if (!near) kept2.push(pg);
    }
    pegs = kept2;
    /* 贴墙壁钉:半嵌在墙里,贴墙球只会被它拨向场内(不会形成夹缝)。
       左墙 face=fieldX0+7,壁钉中心在 face-? 处使球始终位于其场内一侧 */
    var bumpY = pegTop + (rows - 2) * rowGap;   /* 从下往上第二排的高度 */
    pegs.push({ x: fieldX0 + 10, y: bumpY, r: 11, flash: 0, led: null, slideDir: 1 });
    pegs.push({ x: laneX0 - 10, y: bumpY, r: 11, flash: 0, led: null, slideDir: -1 });
    /* ---- 边框 LED 灯带点位(复古主题绘制;极简主题忽略) ---- */
    var leds = [];
    var step = 46;
    for (var lx = cornerL.cx; lx <= cornerR2.cx; lx += step) leds.push({ x: lx, y: ceilingY });
    for (var la = -Math.PI / 2 + 0.4; la < -0.1; la += Math.PI / 9) leds.push({ x: cornerR2.cx
      + Math.cos(la) * (R - 10), y: cornerR2.cy + Math.sin(la) * (R - 10) });
    for (var ly = cornerR2.cy + 30; ly <= floorY - 30; ly += step) leds.push({ x: W - 21, y: ly
       });
    for (var bx = fieldRight - 20; bx >= fieldX0 + 30; bx -= step) leds.push({ x: bx, y: floorY
       + 6 });
    for (var ly2 = floorY - 30; ly2 >= cornerL.cy + 30; ly2 -= step) leds.push({ x: fieldX0, y:
       ly2 });
    for (var la2 = -Math.PI + 0.4; la2 < -Math.PI / 2 - 0.1; la2 += Math.PI / 9) leds.push({ x:
       cornerL.cx + Math.cos(la2) * (R - 10), y: cornerL.cy + Math.sin(la2) * (R - 10) });
    for (var li = 0; li < leds.length; li++) leds[li].color = (li % 2 === 0) ? 'red' : 'green';
    /* 40% 临界速度:按"出生点到顶轨"的爬升高度计算(拉动曲线随屏高自适应) */
    var climb = spawn.y - ceilingY;
    var crestSpeed = Math.sqrt(2 * CFG.gravity * climb + 550 * 550);
    return {
      W: W, H: H,
      crestSpeed: crestSpeed,
      backH: backH, panelH: panelH,
      fieldX0: fieldX0, fieldRight: fieldRight, floorY: floorY,
      lane: { x0: laneX0, x1: laneX1, cx: laneCx, top: L.laneTop, floorY: laneFloorY, spawn: sp
      awn },
      pegs: pegs, bumpers: bumpers,
      holes: holes, pw: pw, capY: capY, holeN: n, innerX0: innerX0,
      dividers: dividers, walls: walls,
      ceilingY: ceilingY, cornerL: cornerL, cornerR: cornerR2,
      borderLeds: leds
    };
  }
  MG.board = { build: build };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
```

```
/* ================================================== */
/* 文件: ./src/game/fx.js */
/* 弹弹珠机 - 粒子与特效 (命名空间: MG.fx) */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  var U = MG.util;
  var MAX_PARTS = 500;
  function create() {
    return { parts: [], rings: [], texts: [], drops: [], shake: 0, sx: 0, sy: 0 };
  }
  /* 弹珠落入洞口/滚回的缩球动画 */
  function drop(fx, x, y, r, color) {
    if (fx.drops.length > 16) fx.drops.shift();
    fx.drops.push({ x: x, y: y, r: r, color: color, age: 0, life: 0.22 });
  }
  function burst(fx, x, y, color, n, spd) {
    spd = spd || 260;
    for (var i = 0; i < n; i++) {
      if (fx.parts.length >= MAX_PARTS) fx.parts.shift();
      var a = U.rand(0, U.TAU);
      var v = U.rand(0.3, 1) * spd;
      fx.parts.push({
        x: x, y: y,
        vx: Math.cos(a) * v, vy: Math.sin(a) * v - 60,
        life: U.rand(0.3, 0.7), age: 0,
        size: U.rand(2.5, 5.5), color: color, grav: 900
      });
    }
  }
  function ring(fx, x, y, color, r1, life, w) {
    fx.rings.push({ x: x, y: y, r0: 6, r1: r1 || 60, age: 0, life: life || 0.35, color: color,
      w: w || 3 });
  }
  function text(fx, x, y, str, opts) {
    opts = opts || {};
    if (fx.texts.length > 24) fx.texts.shift();
    fx.texts.push({
      x: x, y: y, str: str,
      age: 0, life: opts.life || 0.9,
      size: opts.size || 26, color: opts.color || '#ffffff',
      vy: opts.vy != null ? opts.vy : -70
    });
  }
  function shake(fx, p) { fx.shake = Math.min(14, Math.max(fx.shake, p)); }
  function update(fx, dt) {
    var i;
    for (i = fx.parts.length - 1; i >= 0; i--) {
      var p = fx.parts[i];
      p.age += dt;
```

```
      if (p.age >= p.life) { fx.parts.splice(i, 1); continue; }
      p.vy += p.grav * dt;
      p.x += p.vx * dt; p.y += p.vy * dt;
    }
    for (i = fx.rings.length - 1; i >= 0; i--) {
      var r = fx.rings[i];
      r.age += dt;
      if (r.age >= r.life) fx.rings.splice(i, 1);
    }
    for (i = fx.texts.length - 1; i >= 0; i--) {
      var t = fx.texts[i];
      t.age += dt;
      if (t.age >= t.life) { fx.texts.splice(i, 1); continue; }
      t.y += t.vy * dt;
      t.vy *= (1 - 1.6 * dt);
    }
    for (i = fx.drops.length - 1; i >= 0; i--) {
      var dr = fx.drops[i];
      dr.age += dt;
      dr.y += 46 * dt;
      if (dr.age >= dr.life) fx.drops.splice(i, 1);
    }
    if (fx.shake > 0) {
      fx.shake = Math.max(0, fx.shake - dt * 22);
      fx.sx = U.rand(-1, 1) * fx.shake;
      fx.sy = U.rand(-1, 1) * fx.shake;
    } else { fx.sx = 0; fx.sy = 0; }
  }
  function render(ctx, fx) {
    var i;
    ctx.save();
    ctx.globalCompositeOperation = 'lighter';
    for (i = 0; i < fx.parts.length; i++) {
      var p = fx.parts[i];
      var a = 1 - p.age / p.life;
      ctx.globalAlpha = a * 0.9;
      ctx.fillStyle = p.color;
      ctx.beginPath();
      ctx.arc(p.x, p.y, p.size * (0.5 + 0.5 * a), 0, U.TAU);
      ctx.fill();
    }
    for (i = 0; i < fx.rings.length; i++) {
      var r = fx.rings[i];
      var k = r.age / r.life;
      ctx.globalAlpha = (1 - k) * 0.8;
      ctx.strokeStyle = r.color;
      ctx.lineWidth = r.w * (1 - k) + 0.5;
      ctx.beginPath();
      ctx.arc(r.x, r.y, r.r0 + (r.r1 - r.r0) * U.easeOut(k), 0, U.TAU);
      ctx.stroke();
```

```
    }
    for (i = 0; i < fx.drops.length; i++) {
      var dp = fx.drops[i];
      var dk = dp.age / dp.life;
      ctx.globalAlpha = 1 - dk;
      ctx.beginPath();
      ctx.arc(dp.x, dp.y, dp.r * (1 - dk * 0.8), 0, U.TAU);
      ctx.fillStyle = dp.color;
      ctx.fill();
      ctx.globalAlpha = (1 - dk) * 0.6;
      ctx.strokeStyle = 'rgba(90,55,15,0.8)';
      ctx.lineWidth = 1.5;
      ctx.stroke();
    }
    ctx.restore();
    ctx.save();
    ctx.textAlign = 'center';
    ctx.textBaseline = 'middle';
    for (i = 0; i < fx.texts.length; i++) {
      var t = fx.texts[i];
      var k2 = t.age / t.life;
      var pop = t.age < 0.12 ? U.easeOut(t.age / 0.12) * 0.35 + 0.65 : 1;
      ctx.globalAlpha = k2 > 0.6 ? 1 - (k2 - 0.6) / 0.4 : 1;
      ctx.font = 'bold ' + Math.round(t.size * pop) + 'px sans-serif';
      ctx.strokeStyle = 'rgba(0,0,0,0.55)';
      ctx.lineWidth = 4;
      ctx.strokeText(t.str, t.x, t.y);
      ctx.fillStyle = t.color;
      ctx.fillText(t.str, t.x, t.y);
    }
    ctx.restore();
  }
  function reset(fx) {
    fx.parts.length = 0;
    fx.rings.length = 0;
    fx.texts.length = 0;
    fx.shake = 0;
  }
  MG.fx = { create: create, burst: burst, ring: ring, text: text, drop: drop, shake: shake, upd
      ate: update, render: render, reset: reset };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ===================================================== */
/* 文件: ./src/game/ui.js */
/* 弹弹弹珠吧 - UI 绘制与界面 (命名空间: MG.ui)
 * 界面层走 ins 极简风:白卡片 + 墨色胶囊按钮 + 细线 + 柔和投影。
 * 机台本体(黄机台/LED/钉阵)保持街机质感;按钮为立即模式,由游戏层在按下时检测。
 */
(function (root) {
  'use strict';
```

```
var MG = root.MG || (root.MG = {});
var CFG = MG.config;
var U = MG.util;
var C = CFG.colors;
/* 极简 UI 配色 */
var INK = '#3a3a3a';        /* 主文字 / 主按钮 */
var SUB = '#9a9a9a';        /* 次要文字 */
var BODY = '#6a6a6a';       /* 正文 */
var ACCENT = '#e8553f';     /* 强调(数字/关键行) */
var HAIR = 'rgba(0,0,0,0.07)';  /* 发丝描边 */
/* 白卡片:柔和投影 */
function card(ctx, x, y, w, h, r) {
  ctx.save();
  ctx.shadowColor = 'rgba(40,35,25,0.18)';
  ctx.shadowBlur = 34;
  ctx.shadowOffsetY = 10;
  rr(ctx, x, y, w, h, r);
  ctx.fillStyle = 'rgba(255,255,255,0.97)';
  ctx.fill();
  ctx.restore();
  rr(ctx, x, y, w, h, r);
  ctx.strokeStyle = HAIR;
  ctx.lineWidth = 1;
  ctx.stroke();
}
/* ---------- 基础绘制工具 ---------- */
function rr(ctx, x, y, w, h, r) {
  r = Math.min(r, w / 2, h / 2);
  ctx.beginPath();
  ctx.moveTo(x + r, y);
  ctx.arcTo(x + w, y, x + w, y + h, r);
  ctx.arcTo(x + w, y + h, x, y + h, r);
  ctx.arcTo(x, y + h, x, y, r);
  ctx.arcTo(x, y, x + w, y, r);
  ctx.closePath();
}
function font(ctx, size, bold) {
  ctx.font = (bold === false ? '' : 'bold ') + Math.round(size) + 'px sans-serif';
}
function segFont(ctx, size) {
  ctx.font = 'bold ' + Math.round(size) + 'px "Courier New", monospace';
}
function shade(hex, f) {
  var n = parseInt(hex.slice(1), 16);
  var r = Math.round(((n >> 16) & 255) * f);
  var g = Math.round(((n >> 8) & 255) * f);
  var b = Math.round((n & 255) * f);
  return 'rgb(' + r + ',' + g + ',' + b + ')';
}
function rgba(hex, a) {
```

```
  var n = parseInt(hex.slice(1), 16);
  return 'rgba(' + ((n >> 16) & 255) + ',' + ((n >> 8) & 255) + ',' + (n & 255) + ',' + a + '
    )';
}
/* 玻璃弹珠(透明玻璃球 + 色芯 + 高光,暖黄场地上保持高对比) */
function drawMarble(ctx, x, y, r, color) {
  ctx.save();
  /* 投影 */
  ctx.beginPath();
  ctx.ellipse(x + 2, y + 3.5, r * 0.98, r * 0.82, 0, 0, U.TAU);
  ctx.fillStyle = 'rgba(96,58,14,0.25)';
  ctx.fill();
  /* 玻璃体 */
  var g = ctx.createRadialGradient(x - r * 0.32, y - r * 0.38, r * 0.08, x, y, r);
  g.addColorStop(0, '#ffffff');
  g.addColorStop(0.42, rgba(color, 0.96));
  g.addColorStop(0.85, rgba(color, 0.9));
  g.addColorStop(1, 'rgba(255,255,255,0.75)');
  ctx.beginPath();
  ctx.arc(x, y, r, 0, U.TAU);
  ctx.fillStyle = g;
  ctx.fill();
  /* 内部色芯 */
  ctx.beginPath();
  ctx.arc(x + r * 0.12, y + r * 0.14, r * 0.4, 0, U.TAU);
  ctx.fillStyle = rgba(color, 0.72);
  ctx.fill();
  /* 深色轮廓 */
  ctx.strokeStyle = 'rgba(112,74,26,0.6)';
  ctx.lineWidth = 1.6;
  ctx.beginPath();
  ctx.arc(x, y, r, 0, U.TAU);
  ctx.stroke();
  /* 高光 + 底部反光 */
  ctx.beginPath();
  ctx.arc(x - r * 0.36, y - r * 0.42, r * 0.24, 0, U.TAU);
  ctx.fillStyle = 'rgba(255,255,255,0.95)';
  ctx.fill();
  ctx.beginPath();
  ctx.arc(x + r * 0.3, y + r * 0.42, r * 0.1, 0, U.TAU);
  ctx.fillStyle = 'rgba(255,255,255,0.6)';
  ctx.fill();
  ctx.restore();
}
/* 五角星贴纸 */
function drawStarSticker(ctx, x, y, R, rot, alpha) {
  ctx.save();
  ctx.globalAlpha = alpha == null ? 1 : alpha;
  ctx.beginPath();
  for (var i = 0; i < 10; i++) {
```

```
    ctx.fillStyle = 'rgba(255,255,255,0.85)';
    var ft = '玩具球 ' + game.prizeBallsTotal + '    礼品卡 ' + game.giftCardsTotal + '    最大单次 +'
+
      U.fmt(game.bestWin);
    ctx.fillText(ft, W / 2, bottom - 166);
    /* ---- 审核要求:健康游戏忠告(官方原文) + 适龄提示(官方 CADPA 标识) ---- */
    font(ctx, 13);
    ctx.fillStyle = 'rgba(255,255,255,0.92)';
    ctx.fillText('《健康游戏忠告》', W / 2, bottom - 146);
    font(ctx, 12, false);
    ctx.fillStyle = 'rgba(255,255,255,0.72)';
    ctx.fillText('抵制不良游戏，拒绝盗版游戏。注意自我保护，谨防受骗上当。', W / 2, bottom - 122);
    ctx.fillText('适度游戏益脑，沉迷游戏伤身。合理安排时间，享受健康生活。', W / 2, bottom - 100);
    var badgeY = bottom - 52;
    /* 三级加载:文件路径 → 内嵌 data URI → 手绘兜底 */
    if (!game.ageBadge && MG.ageBadgeData && !game.ageBadgeFailed) {
      try {
        var bi = game.platform.createImage();
        var failToData = function () {
          bi.onload = function () { game.ageBadge = bi; };
          bi.onerror = function () { game.ageBadgeFailed = true; };
          bi.src = MG.ageBadgeData;
        };
        bi.onload = function () { game.ageBadge = bi; };
        bi.onerror = failToData;
        bi.src = 'assets/logo/cadpa-8.png';
      } catch (e) { game.ageBadgeFailed = true; }
    }
    if (game.ageBadge) {
      var bh2 = 64, bw2 = bh2 * (game.ageBadge.width / game.ageBadge.height);
      ctx.drawImage(game.ageBadge, W / 2 - 250, badgeY - bh2 / 2, bw2, bh2);
    } else {
      rr(ctx, W / 2 - 250, badgeY - 17, 40, 34, 8);
      ctx.fillStyle = '#2fa84f';
      ctx.fill();
      font(ctx, 18);
      ctx.fillStyle = '#ffffff';
      ctx.fillText('8+', W / 2 - 230, badgeY);
    }
    ctx.textAlign = 'left';
    font(ctx, 13, false);
    ctx.fillStyle = 'rgba(255,255,255,0.88)';
    ctx.fillText('适龄提示：本游戏适合8周岁以上用户使用', W / 2 - 150, badgeY);
  }
  /* ---------- 玩法说明 ---------- */
  function renderHelp(game) {
    var ctx = game.ctx;
    var W = game.view.W, H = game.view.H;
    ctx.fillStyle = 'rgba(25,22,18,0.5)';
    ctx.fillRect(0, 0, W, H);
    var lines = [
```

```
      '右下角先选投入弹珠数:1~5 颗',
      '投入越多倍率越高:×2 → ×3 → ×4 → ×5 → ×6',
      '倍率越高,底部亮灯的洞越少(×6 只亮 1 个)',
      '按住蓄力,松手把弹珠从右下角往上弹(点按只是轻弹)',
      '力度太小弹珠会滚回来,退还投入',
      '弹珠从亮灯的洞滑落 = 中奖,赢回 投入×倍率 颗',
      '每次中奖送玩具球,×6 以上再送礼品卡;弹珠库见底则结束'
    ];
    var cw = Math.min(690, W - 40);
    var ch = 90 + lines.length * 56 + 120;
    var cx = W / 2 - cw / 2, cy = H / 2 - ch / 2;
    card(ctx, cx, cy, cw, ch, 30);
    ctx.textAlign = 'center';
    ctx.textBaseline = 'middle';
    font(ctx, 36);
    ctx.fillStyle = INK;
    ctx.fillText('玩法说明', W / 2, cy + 56);
    font(ctx, 24, false);
    for (var i = 0; i < lines.length; i++) {
      ctx.fillStyle = i === 4 ? ACCENT : BODY;
      ctx.fillText(lines[i], W / 2, cy + 118 + i * 52);
    }
    var bw = 300, bh = 74;
    button(game, W / 2 - bw / 2, cy + ch - bh - 30, bw, bh, '返回', function () {
      game.state = 'menu'; game.stateT = 0;
    }, { primary: true, size: 30 });
  }
  /* ---------- 暂停 ---------- */
  function renderPause(game) {
    var ctx = game.ctx;
    var W = game.view.W, H = game.view.H;
    var arcade = CFG.themeName === 'arcade';
    ctx.fillStyle = arcade ? 'rgba(50,30,5,0.6)' : 'rgba(25,22,18,0.45)';
    ctx.fillRect(0, 0, W, H);
    var cw = 520, ch = 508;
    var cx = W / 2 - cw / 2, cy = H / 2 - ch / 2;
    if (arcade) {
      rr(ctx, cx, cy, cw, ch, 26);
      ctx.fillStyle = C.cream;
      ctx.fill();
      ctx.strokeStyle = '#e8b45a';
      ctx.lineWidth = 3;
      ctx.stroke();
    } else {
      card(ctx, cx, cy, cw, ch, 30);
    }
    ctx.textAlign = 'center';
    ctx.textBaseline = 'middle';
    font(ctx, 36);
    ctx.fillStyle = arcade ? '#6b4409' : INK;
```

```
ctx.fillText('已暂停', W / 2, cy + 56);
var bw = 400, bh = 66, bx = W / 2 - bw / 2;
button(game, bx, cy + 96, bw, bh, '继续游戏', function () { game.state = 'play'; game.stateT =
  0; MG.music.play(); }, { primary: true, size: 30 });
button(game, bx, cy + 172, bw, bh, '重新开始', function () { game.restartRound(); }, { size: 28
   });
button(game, bx, cy + 242, bw, 48, '机台风格 · ' + game.themeLabel(), function () {
  game.setTheme(game.theme === 'arcade' ? 'min' : 'arcade');
}, { size: 24, bold: false });
/* 双声道音量滑条:音效 / 音乐 各自独立 */
var tx = cx + 60, tw = cw - 120;
game.sliders = [
  { key: 'sfx', x: tx, w: tw, knobY: cy + 330 },
  { key: 'music', x: tx, w: tw, knobY: cy + 398 }
];
var rows = [
  { sl: game.sliders[0], label: '音效', v: game.volSfx },
  { sl: game.sliders[1], label: '音乐', v: game.volMusic }
];
for (var si = 0; si < rows.length; si++) {
  var row = rows[si];
  var trackY = row.sl.knobY;
  rr(ctx, tx, trackY - 6, tw, 12, 6);
  ctx.fillStyle = arcade ? 'rgba(90,55,15,0.25)' : 'rgba(0,0,0,0.08)';
  ctx.fill();
  var fv = Math.max(0, Math.min(1, row.v || 0));
  var fillW = Math.round(tw * fv);
  if (fillW > 6) {
    rr(ctx, tx, trackY - 6, fillW, 12, 6);
    ctx.fillStyle = arcade ? '#a4622a' : INK;
    ctx.fill();
  }
  if (arcade) {
    ctx.beginPath();
    ctx.arc(tx + fillW, trackY, 14, 0, U.TAU);
    ctx.fillStyle = '#fff8ea';
    ctx.fill();
    ctx.strokeStyle = '#a4622a';
    ctx.lineWidth = 3;
    ctx.stroke();
  } else {
    ctx.save();
    ctx.shadowColor = 'rgba(40,35,25,0.25)';
    ctx.shadowBlur = 8;
    ctx.shadowOffsetY = 2;
    ctx.beginPath();
    ctx.arc(tx + fillW, trackY, 14, 0, U.TAU);
    ctx.fillStyle = '#ffffff';
    ctx.fill();
    ctx.restore();
```

```
        ctx.beginPath();
        ctx.arc(tx + fillW, trackY, 14, 0, U.TAU);
        ctx.strokeStyle = HAIR;
        ctx.lineWidth = 1;
        ctx.stroke();
      }
      ctx.textAlign = 'left';
      font(ctx, 19, false);
      ctx.fillStyle = arcade ? C.panelText : SUB;
      ctx.fillText(row.label, tx, trackY - 26);
      ctx.textAlign = 'right';
      ctx.fillText(Math.round(fv * 100) + '%', tx + tw, trackY - 26);
    }
  }
  /* ---------- 结算 ---------- */
  function renderResult(game) {
    var ctx = game.ctx;
    var W = game.view.W, H = game.view.H;
    ctx.fillStyle = 'rgba(25,22,18,0.5)';
    ctx.fillRect(0, 0, W, H);
    var compact = H < 1400;
    var cw = 620, ch = compact ? 540 : 600;
    var cx = W / 2 - cw / 2, cy = H / 2 - ch / 2;
    card(ctx, cx, cy, cw, ch, 30);
    ctx.textAlign = 'center';
    ctx.textBaseline = 'middle';
    font(ctx, 38);
    ctx.fillStyle = INK;
    ctx.fillText('弹珠用完啦', W / 2, cy + 78);
    font(ctx, 24, false);
    ctx.fillStyle = SUB;
    ctx.fillText('发射 ' + game.launched + ' 次 · 中奖 ' + game.wins + ' 次', W / 2, cy + 142);
    font(ctx, 62);
    ctx.fillStyle = ACCENT;
    ctx.fillText('最大单次 +' + U.fmt(game.lastEarned), W / 2, cy + 208);
    font(ctx, 26, false);
    ctx.fillStyle = BODY;
    ctx.fillText('玩具球 ×' + game.prizeBalls + '    礼品卡 ×' + game.giftCards, W / 2, cy + 282);
    font(ctx, 25, false);
    ctx.fillStyle = SUB;
    ctx.fillText('最大单次纪录  +' + U.fmt(game.bestWin), W / 2, cy + 336);
    if (game.newRecord) {
      var bw2 = 110, bh2 = 40;
      rr(ctx, W / 2 + 120, cy + 336 - bh2 / 2, bw2, bh2, bh2 / 2);
      ctx.fillStyle = ACCENT;
      ctx.fill();
      font(ctx, 22, false);
      ctx.fillStyle = '#fff';
      ctx.fillText('新纪录', W / 2 + 175, cy + 336);
    }
```

```
    var bw = 420, bh = 82, bx = W / 2 - bw / 2;
    var by = cy + (compact ? 400 : 430);
    if (game.platform.isMinigame) {
      button(game, bx, by, bw, bh * 0.72, '分享给好友', function () {
        game.platform.share('我在「弹弹弹珠吧」中了 ' + game.wins + ' 次,最大单次赢 ' + U.fmt(game.maxWi
n) + ' 颗
      弹珠!');
      }, { size: 26 });
      by += bh * 0.72 + 24;
    }
    button(game, bx, by, bw, bh, '再来一局(补20颗)', function () { game.restartRound(); }, { primary:
       true, size: 32 });
  }
  MG.ui = {
    rr: rr,
    font: font,
    segFont: segFont,
    shade: shade,
    rgba: rgba,
    drawMarble: drawMarble,
    drawStarSticker: drawStarSticker,
    drawPuppy: drawPuppy,
    button: button,
    iconButton: iconButton,
    renderMenu: renderMenu,
    renderHelp: renderHelp,
    renderPause: renderPause,
    renderResult: renderResult
  };
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ================================================== */
/* 文件: ./src/game/marble-game.js */
/* 弹弹弹珠吧 - 游戏主逻辑 (命名空间: MG.Game)
 * 复刻玻璃珠游艺机:
 *   - 右下角先选投入弹珠数(1~5 颗)= 选倍率 ×2~×10,投入越多亮灯的洞越少
 *   - 按住蓄力,松手把弹珠从右下角往上弹,冲过顶部导弧拐入斜面场地
 *   - 力度太小弹珠滚回来,退还投入;从亮灯的洞滑落 = 中奖,赢回 投入×倍率 颗
 *   - ×6 及以上中奖送礼品卡,每次中奖送玩具球;弹珠库见底本局结束
 * 状态机: menu / help / play / pause / result
 */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  var CFG = MG.config;
  var U = MG.util;
  var C = CFG.colors;
  function Game(platform) {
    var self = this;
    this.platform = platform;
    this.ctx = platform.canvas.getContext('2d');
```

```
    /* 存档 */
    this.muted = !!platform.storage.get('mm_mute', false);
    MG.audio.init(platform);
    MG.audio.setMuted(this.muted);
    this.prizeBallsTotal = platform.storage.get('mm_prizes', 0) | 0;
    this.giftCardsTotal = platform.storage.get('mm_cards', 0) | 0;
    this.bestWin = platform.storage.get('mm_bestwin', 0) | 0;
    this.songIdx = platform.storage.get('mm_song', 0) | 0;
    /* 双声道音量:音效 / 音乐各自独立(旧 mm_vol 迁移为两者初值) */
    var legacyVol = Math.max(0, Math.min(1, platform.storage.get('mm_vol', 0.8)));
    this.volSfx = Math.max(0, Math.min(1, platform.storage.get('mm_vol_sfx', legacyVol)));
    this.volMusic = Math.max(0, Math.min(1, platform.storage.get('mm_vol_music', legacyVol)));
    MG.music.init(platform);
    MG.music.setMuted(this.muted);   /* 同步存档静音态:静音时解锁手势后不得自启 BGM */
    MG.music.setVolume(this.volMusic);
    MG.audio.setVolume(this.volSfx);
    MG.music.setSong(this.songIdx);
    /* 视口 */
    this.view = { scale: 1, offX: 0, offY: 0, W: CFG.DESIGN_W, H: 1500 };
    this.computeView();
    /* 状态 */
    this.state = 'menu';
    this.stateT = 0;
    this.time = 0;
    this.bank = 0;              // 弹珠库
    this.selIdx = 0;            // 选中的投入档位
    this.launched = 0;          // 本局发射次数
    this.wins = 0;
    this.maxWin = 0;
    this.prizeBalls = 0;
    this.giftCards = 0;
    this.tierPopT = 0;
    this.newRecord = false;
    this.lastEarned = 0;        // 结算页:最大单次赢取
    this.skinIdx = 0;
    this.endTimer = -1;
    this.launcherCool = 0;
    this.launcherPulse = 0;
    this.charging = false;
    this.chargeT = 0;
    this.sliderArmed = false;
    this._downT = 0;
    this.frameButtons = [];
    this.fx = MG.fx.create();
    this.bg = null;
    /* 主题:极简(默认)/ 复古街机 */
    this.theme = platform.storage.get('mm_theme', 'min') === 'arcade' ? 'arcade' : 'min';
    CFG.setTheme(this.theme);
    this.buildBoard(0);
    /* 输入 */
```

```
  platform.onTouch('down', function (pt) { self.handleDown(pt); });
  platform.onTouch('move', function (pt) {
    if (self.sliderArmed) {
      var p = self.toLogical(pt.x, pt.y);
      self.setChannelVolume(self.sliderArmed, p.x);
    }
  });
  platform.onTouch('up', function (pt) { self.handleUp(pt); });
  platform.onResize(function () {
    self.computeView();
    self.buildBoard(self.round || 0);
  });
}
/* ================ 视口 ================ */
Game.prototype.computeView = function () {
  var p = this.platform;
  var safe = p.safeTop || 0;                  /* 状态栏+胶囊占用的顶部安全区 */
  var availH = Math.max(400, p.cssH - safe);
  var H = U.clamp(CFG.DESIGN_W * availH / p.cssW, CFG.H_MIN, CFG.H_MAX);
  var scale = Math.min(p.cssW / CFG.DESIGN_W, availH / H);
  this.view.W = CFG.DESIGN_W;
  this.view.H = H;
  this.view.scale = scale;
  this.view.offX = (p.cssW - CFG.DESIGN_W * scale) / 2;
  this.view.offY = safe + (availH - H * scale) / 2;
  /* 顶部图标簇的高度:小游戏端对齐胶囊按钮中心,浏览器端固定 42 */
  this.iconRowY = 42;
  if (p.capsuleY) {
    this.iconRowY = U.clamp((p.capsuleY + (p.iconRowAdjCss || 0) - this.view.offY) / scale, 8
    , 96);
  }
};
Game.prototype.toLogical = function (cx, cy) {
  return {
    x: (cx - this.view.offX) / this.view.scale,
    y: (cy - this.view.offY) / this.view.scale
  };
};
/* ================ 关卡构建 ================ */
Game.prototype.buildBoard = function (round) {
  var seed = round > 0 ? ((round * 2654435761) >>> 0) : 20260101;
  var rng = U.mulberry32(seed);
  this.round = round;
  this.board = MG.board.build(round, this.view.W, this.view.H, rng);
  this.world = new MG.physics.World({
    gravity: CFG.gravity, maxSpeed: CFG.maxSpeed, airDrag: CFG.airDrag
  });
  var i;
  for (i = 0; i < this.board.pegs.length; i++) {
    var peg = this.board.pegs[i];
```

```
      peg.kind = 'peg'; peg.e = 0.12;   /* 钉子要"拨"不要"弹":撞后仅回弹 12% */
      this.world.addCircle(peg);
    }
    for (i = 0; i < this.board.bumpers.length; i++) {
      var bumper = this.board.bumpers[i];
      bumper.e = 0.25;   /* 大弹柱调低:不再当炮台,只拨球 */
      this.world.addCircle(bumper);
    }
    for (i = 0; i < this.board.dividers.length; i++) this.world.addSeg(this.board.dividers[i]);
    for (i = 0; i < this.board.walls.length; i++) this.world.addSeg(this.board.walls[i]);
    var self = this;
    this.world.onEvent = function (e) { self.onWorldEvent(e); };
    this.endTimer = -1;
    this.launcherCool = 0;
    this.charging = false;
    this.buildBg();
  };
  /* ================ 流程 ================ */
  Game.prototype.startGame = function () {
    this.bank = CFG.startBank;
    this.selIdx = 0;
    this.launched = 0;
    this.wins = 0;
    this.maxWin = 0;
    this.prizeBalls = 0;
    this.giftCards = 0;
    this.buildBoard(1);
    this.applyLit();
    this.state = 'play';
    this.stateT = 0;
    MG.audio.play('click');
  };
  Game.prototype.restartRound = function () {
    MG.music.play();   /* 从暂停恢复时音乐接着放 */
    this.bank = CFG.startBank;
    this.selIdx = 0;
    this.launched = 0;
    this.wins = 0;
    this.maxWin = 0;
    this.prizeBalls = 0;
    this.giftCards = 0;
    this.buildBoard(this.round);
    this.applyLit();
    this.state = 'play';
    this.stateT = 0;
    MG.audio.play('click');
  };
  Game.prototype.backToMenu = function () {
    this.state = 'menu';
    this.stateT = 0;
```

```
  this.buildBoard(0);
  MG.audio.play('click');
};
Game.prototype.cycleMusic = function () {
  var name = MG.music.next();
  this.platform.storage.set('mm_song', MG.music.current());
  MG.fx.text(this.fx, 106, 96, '♪ ' + name, { size: 24, color: '#fff8ea', life: 1.5 });
  MG.audio.play('click');
};
Game.prototype.toggleMute = function () {
  this.muted = MG.audio.toggleMuted();
  this.platform.storage.set('mm_mute', this.muted);
  MG.music.setMuted(this.muted);
  if (!this.muted) MG.audio.play('click');
};
Game.prototype.endRound = function () {
  this.platform.storage.set('mm_prizes', this.prizeBallsTotal + this.prizeBalls);
  this.platform.storage.set('mm_cards', this.giftCardsTotal + this.giftCards);
  this.newRecord = this.maxWin > this.bestWin && this.maxWin > 0;
  if (this.newRecord) {
    this.bestWin = this.maxWin;
    this.platform.storage.set('mm_bestwin', this.bestWin);
  }
  this.lastEarned = this.maxWin;
  this.prizeBallsTotal += this.prizeBalls;
  this.giftCardsTotal += this.giftCards;
  this.state = 'result';
  this.stateT = 0;
  MG.audio.play(this.newRecord ? 'record' : 'over');
};
/* ================= 倍率档位与亮灯 ================= */
Game.prototype.tier = function () {
  return CFG.tiers[U.clamp(this.selIdx, 0, CFG.tiers.length - 1)];
};
/* 右下角选投入弹珠数 */
Game.prototype.selectTier = function (idx) {
  var t = CFG.tiers[idx];
  if (!t || t.cost > this.bank || idx === this.selIdx) return;
  this.selIdx = idx;
  this.applyLit();
  this.tierPopT = 0.2;
  MG.audio.play('click');
};
/* 按当前档位随机点亮洞口 */
Game.prototype.applyLit = function () {
  var holes = this.board.holes;
  var i;
  for (i = 0; i < holes.length; i++) holes[i].lit = false;
  var order = [];
  for (i = 0; i < holes.length; i++) order.push(i);
```

```
  for (i = order.length - 1; i > 0; i--) {
    var k = Math.floor(Math.random() * (i + 1));
    var tmp = order[i]; order[i] = order[k]; order[k] = tmp;
  }
  var litN = Math.min(this.tier().lit, holes.length);
  for (i = 0; i < litN; i++) holes[order[i]].lit = true;
};
/* ================ 输入:按住蓄力 / 松手发射 ================ */
Game.prototype.handleDown = function (pt) {
  MG.audio.unlock();
  MG.music.unlock();
  var p = this.toLogical(pt.x, pt.y);
  for (var i = this.frameButtons.length - 1; i >= 0; i--) {
    var b = this.frameButtons[i];
    if (p.x >= b.x && p.x <= b.x + b.w && p.y >= b.y && p.y <= b.y + b.h) {
      b.cb();
      return;
    }
  }
  if (this.state === 'pause' && this.sliders) {
    for (var s = 0; s < this.sliders.length; s++) {
      var sl = this.sliders[s];
      if (p.x >= sl.x - 20 && p.x <= sl.x + sl.w + 20 && Math.abs(p.y - sl.knobY) < 32) {
        this.sliderArmed = sl;
        this.setChannelVolume(sl, p.x);
        return;
      }
    }
  }
  if (this.state !== 'play') return;
  this.charging = true;
  this.chargeT = 0;
  this._downT = this.time;
};
/* 分声道音量:sl = renderPause 注册的滑条描述 */
Game.prototype.setChannelVolume = function (sl, x) {
  var v = U.clamp((x - sl.x) / sl.w, 0, 1);
  v = Math.round(v * 100) / 100;
  if (sl.key === 'music') {
    this.volMusic = v;
    MG.music.setVolume(v);
    this.platform.storage.set('mm_vol_music', v);
  } else {
    this.volSfx = v;
    MG.audio.setVolume(v);
    this.platform.storage.set('mm_vol_sfx', v);
  }
};
Game.prototype.handleUp = function () {
  if (this.sliderArmed) { this.sliderArmed = false; return; }
```

```
  if (!this.charging) return;
  this.charging = false;
  if (this.state !== 'play') return;
  var quick = (this.time - this._downT) < CFG.launcher.tapJudge;
  /* 点按绕过非线性曲线,直接用原始力度(5% 轻弹) */
  var power = quick ? CFG.launcher.tapPower : this.chargePower();
  this.launch(power);
};
/* 力度三角波:0 → 1 → 0 来回摆动 */
Game.prototype.chargePower = function () {
  var c = (this.chargeT * CFG.launcher.chargeSpeed) % 1;
  return c < 0.5 ? c * 2 : 2 - c * 2;
};
/* 发射:右下角向上弹(场上同时只能有一颗弹珠) */
Game.prototype.launch = function (power) {
  if (this.state !== 'play' || this.launcherCool > 0) return;
  if (this.world.balls.length > 0) {
    this.fxLabel(this.board.lane.cx - 10, this.board.lane.floorY - 130, '等上一颗落定', { size: 22,
     color: '#fff8ea', life: 0.8 });
    return;
  }
  var tier = this.tier();
  if (this.bank < tier.cost) {
    this.fxLabel(this.board.lane.cx - 10, this.board.lane.floorY - 80, '弹珠不够', { size: 22, co
    lor: '#e8352e' });
    return;
  }
  this.bank -= tier.cost;
  this.launched++;
  /* 亮灯洞只在换档位时重排,发射前后保持不变 */
  var lane = this.board.lane;
  /* 两段式力度:前 40% 曲线(到临界),后 60% 线性到最大 */
  var cp = CFG.launcher.crestPower;
  var crest = this.board.crestSpeed || CFG.launcher.crestSpeed;   /* 随屏高自适应 */
  var frac = (crest - CFG.launcher.minSpeed) / (CFG.launcher.maxSpeed - CFG.launcher.minSpeed
    );
  var lp;
  if (power <= cp) {
    var t = power / cp;
    t = t * t * (3 - 2 * t);   /* smoothstep 曲线 */
    lp = t * frac;
  } else {
    lp = frac + (1 - frac) * ((power - cp) / (1 - cp));
  }
  var speed = U.lerp(CFG.launcher.minSpeed, CFG.launcher.maxSpeed, lp);
  /* 垂直上升:无水平抖动 */
  /* 出膛随机抖动:真机弹珠在弹道内颤动,混沌放大让每次落点不同 */
  var jx = U.rand(-16, 16);
  var m = new MG.entities.Marble(lane.spawn.x + jx * 0.4, lane.spawn.y, jx * 3, -speed * U.ra
    nd(0.975, 1.025), this.skinIdx++ % C.marbleTints.length);
```

```
  m.bet = tier.cost;
  m.mult = tier.mult;
  this.world.addBall(m);
  this.launcherCool = CFG.launcher.cooldown;
  this.launcherPulse = 1;
  MG.audio.play('launch');
  MG.fx.burst(this.fx, lane.spawn.x, lane.spawn.y, '#ffd166', 10, 240);
};
/* ================= 物理事件 ================= */
Game.prototype.onWorldEvent = function (e) {
  switch (e.type) {
    case 'peg': {
      e.circle.flash = 1;
      MG.fx.burst(this.fx, e.x, e.y, e.circle.led === 'red' ? '#ff8a7a' : (e.circle.led === '
    green' ? '#8affb0' : '#ffffff'), 4, 130);
      if (e.speed > 70) MG.audio.play('peg', { pitch: U.rand(0.9, 1.2) });
      break;
    }
    case 'bumper': {
      e.circle.pulse = 1;
      MG.fx.ring(this.fx, e.x, e.y, '#ffffff', 54, 0.3, 3);
      if (!(e.circle.sndCd > 0)) {
        e.circle.sndCd = 0.15;   /* 同柱 0.15s 内不重复发声 */
        MG.audio.play('bumper', { vol: Math.min(1, 0.4 + e.speed / 2000) });
      }
      break;
    }
    case 'wall': {
      if (e.ball.wallSndT <= 0 && e.speed > 420) {
        e.ball.wallSndT = 0.2;
        MG.audio.play('peg', { pitch: 0.55, vol: 0.3 });
      }
      break;
    }
  }
};
/* 居中特效文字并夹在机台内(中文按约 0.92em 估算宽度,防两端被裁) */
Game.prototype.fxLabel = function (x, y, str, opts) {
  opts = opts || {};
  var size = opts.size || 24;
  var half = str.length * size * 0.46 + 14;
  var cx = U.clamp(x, half, this.view.W - half);
  MG.fx.text(this.fx, cx, y, str, opts);
};
/* 弹道内滚回:退还投入 */
Game.prototype.refund = function (m) {
  this.bank += (m.bet || 1);
  MG.fx.drop(this.fx, m.x, m.y, m.r, C.marbleTints[m.skin % C.marbleTints.length]);
  m.dead = true;
  var lane = this.board.lane;
```

```
  this.fxLabel(lane.cx - 20, lane.floorY - 90, '力气太小 退' + (m.bet || 1) + '颗', { size: 24, col
    or: '#fff8ea', life: 1.1 });
  MG.audio.play('click');
};
/* 卡住解救失败:退回投入收球(不按下方洞口结算,卡在高处也不白给) */
Game.prototype.rescueRefund = function (m) {
  this.bank += (m.bet || 1);
  MG.fx.drop(this.fx, m.x, m.y, m.r, C.marbleTints[m.skin % C.marbleTints.length]);
  m.dead = true;
  MG.fx.text(this.fx, m.x, m.y - 44, '退' + (m.bet || 1) + '颗', { size: 22, color: '#9a9a9a',
    life: 1.1 });
  MG.audio.play('click');
};
/* 切换机台主题(极简 / 复古街机):只重绘外观缓存,
   不动棋盘/物理世界——在飞的弹珠与亮灯洞口原样保留 */
Game.prototype.setTheme = function (t) {
  this.theme = (t === 'arcade') ? 'arcade' : 'min';
  this.platform.storage.set('mm_theme', this.theme);
  MG.config.setTheme(this.theme);
  this.buildBg();
};
Game.prototype.themeLabel = function () {
  return this.theme === 'arcade' ? '复古街机' : '极简';
};
/* 落入洞口:亮灯 = 中奖 */
Game.prototype.captureHole = function (m) {
  var b = this.board;
  var idx = U.clamp(Math.floor((m.x - b.innerX0) / b.pw), 0, b.holeN - 1);
  var hole = b.holes[idx];
  m.dead = true;
  if (hole.lit) {
    var gained = (m.bet || 1) * (m.mult || this.tier().mult);
    this.bank += gained;
    this.wins++;
    if (gained > this.maxWin) this.maxWin = gained;
    this.prizeBalls += 1;
    var gotCard = (m.mult || 2) >= CFG.giftCardMult;
    if (gotCard) this.giftCards += 1;
    hole.flash = 1;
    MG.fx.drop(this.fx, m.x, m.y, m.r, C.marbleTints[m.skin % C.marbleTints.length]);
    MG.fx.text(this.fx, this.view.W / 2, b.floorY - 170, '中 奖  ×' + (m.mult || 2), { size: 56
    , color: '#e8352e', life: 1.6 });
    this.fxLabel(hole.cx, b.floorY - 96, '+' + gained + ' 颗  玩具球+1' + (gotCard ? '  礼品卡+1' :
    ''), { size: 26, color: '#fff8ea', life: 1.5 });
    MG.fx.burst(this.fx, hole.cx, b.floorY - 30, '#ff4a3a', 44, 440);
    MG.fx.ring(this.fx, hole.cx, b.floorY - 30, '#ff4a3a', 120, 0.5, 6);
    MG.audio.play('jackpot');
    MG.fx.shake(this.fx, 10);
    this.platform.vibrate();
  } else {
```

```
      hole.flash = 0.6;
      MG.fx.drop(this.fx, m.x, m.y, m.r, C.marbleTints[m.skin % C.marbleTints.length]);
      MG.fx.burst(this.fx, m.x, b.floorY - 26, '#d9b26a', 8, 180);
      MG.audio.play('slot', { pitch: 0.85, vol: 0.6 });
    }
    /* 每颗弹珠落定后重置亮灯洞口位置(发射前后不变) */
    this.applyLit();
  };
  /* ================= 更新 ================= */
  Game.prototype.update = function (dt) {
    this.time += dt;
    this.stateT += dt;
    this.tierPopT = Math.max(0, this.tierPopT - dt);
    MG.fx.update(this.fx, dt);
    if (this.state !== 'play') return;
    var b = this.board;
    var i, k;
    /* 蓄力:场上还有球或冷却中时,力度条冻结(松手即所见力度) */
    if (this.charging && this.world.balls.length === 0 && this.launcherCool <= 0) {
      this.chargeT += dt;
    }
    this.launcherCool -= dt;
    this.launcherPulse = Math.max(0, this.launcherPulse - dt * 4);
    this.world.step(dt, 12);
    /* 弹珠簿记 */
    var balls = this.world.balls;
    var alive = 0;
    for (i = 0; i < balls.length; i++) {
      var m = balls[i];
      if (m.dead) continue;
      alive++;
      m.age += dt;
      m.wallSndT -= dt;
      /* 拖尾 */
      m.trailT -= dt;
      if (m.trailT <= 0) {
        m.trailT = 0.035;
        m.trail.push({ x: m.x, y: m.y });
        if (m.trail.length > 7) m.trail.shift();
      }
      var speed = Math.sqrt(m.vx * m.vx + m.vy * m.vy);
      /* 弹道内滚回检测 */
      if (m.x > b.lane.x0 - 5 && m.y > b.lane.floorY - 190) {
        if (speed < CFG.launcher.refundSlow) m.refundT = (m.refundT || 0) + dt;
        else m.refundT = 0;
        if (m.refundT > CFG.launcher.refundTime) { this.refund(m); continue; }
      } else {
        m.refundT = 0;
        /* 落入洞口:球必须完全落进洞口内(帽线以下、两针之间)才计算,
           在隔针顶/侧面弹跳一律不算 */
```

```
      if (m.y > b.capY - 6 && m.y < b.floorY + 30 && m.x < b.lane.x0 - 10) {
        this.captureHole(m);
        continue;
      }
      /* 卡住解救(通用,不指定落洞):持续低速 → 往下抖落;
         反复解救无效 → 退还投入收球。卡在高处也不会按下方洞口白给 */
      if (speed < 40) m.stuckTotal = (m.stuckTotal || 0) + dt;
      else m.stuckTotal = Math.max(0, (m.stuckTotal || 0) - dt * 2);
      if (m.rescueCd > 0) m.rescueCd -= dt;
      if (m.stuckTotal > 2.2 && !(m.rescueCd > 0)) {
        m.rescueCd = 0.6;
        m.rescues = (m.rescues || 0) + 1;
        m.vx += U.rand(-240, 240);
        m.vy += U.rand(60, 150);   /* 只往下抖,不往高处跳 */
        MG.fx.text(this.fx, m.x, m.y - 26, '解救', { size: 18, color: '#9a9a9a', life: 0.7 });
      }
      if (m.rescues > 4 || m.age > 25) {
        this.rescueRefund(m);
        continue;
      }
    }
  }
  /* 闪光衰减 */
  for (i = 0; i < b.pegs.length; i++) b.pegs[i].flash = Math.max(0, b.pegs[i].flash - dt * 3.
    5);
  for (i = 0; i < b.bumpers.length; i++) {
    b.bumpers[i].pulse = Math.max(0, b.bumpers[i].pulse - dt * 3);
    if (b.bumpers[i].sndCd > 0) b.bumpers[i].sndCd -= dt;
  }
  for (i = 0; i < b.holes.length; i++) b.holes[i].flash = Math.max(0, b.holes[i].flash - dt *
     2);
  /* 弹珠库见底 → 本局结束 */
  if (this.bank < 1 && alive === 0) {
    if (this.endTimer < 0) this.endTimer = 1.0;
    this.endTimer -= dt;
    if (this.endTimer <= 0) { this.endTimer = -1; this.endRound(); }
  }
};
/* ================= 背景缓存(整个机台) ================= */
Game.prototype.buildBg = function () {
  var p = this.platform;
  var W = this.view.W, H = this.view.H;
  var b = this.board;
  var off = p.createOffscreen(p.canvas.width, p.canvas.height);
  var ctx = off.getContext('2d');
  var ui = MG.ui;
  if (p.isMinigame) {                       /* 安全区/留边铺机身底色 */
    ctx.fillStyle = C.cabYellowHi;
    ctx.fillRect(0, 0, off.width, off.height);
  }
```

```
    var s = this.view.scale * p.dpr;
    ctx.setTransform(s, 0, 0, s, this.view.offX * p.dpr, this.view.offY * p.dpr);
    /* ---- 机身(金黄) ---- */
    ui.rr(ctx, -3, -3, W + 6, H + 6, 34);
    var body = ctx.createLinearGradient(0, 0, 0, H);
    body.addColorStop(0, C.cabYellowHi);
    body.addColorStop(0.5, C.cabYellow);
    body.addColorStop(1, C.cabYellowLo);
    ctx.fillStyle = body;
    ctx.fill();
    /* 白色外框 */
    ui.rr(ctx, 8, 8, W - 16, H - 16, 28);
    ctx.strokeStyle = C.frame;
    ctx.lineWidth = 12;
    ctx.stroke();
    var ARC = CFG.themeName === 'arcade';
    /* ---- 顶部标题面板(直接印在机身,上接白框、下接顶轨,四边齐平不浮空) ---- */
    ctx.textAlign = 'center';
    ctx.textBaseline = 'middle';
    if (ARC) {
      ui.drawStarSticker(ctx, 60, 120, 13, 0.2, 0.8);
      ui.drawStarSticker(ctx, 268, 44, 10, 0.5, 0.7);
      ui.drawStarSticker(ctx, 600, 40, 12, 0.1, 0.8);
      ui.drawStarSticker(ctx, 40, 240, 11, 0.7, 0.7);
      ui.drawStarSticker(ctx, 300, 250, 9, 0.3, 0.6);
      ui.drawPuppy(ctx, 152, (this.iconRowY || 42) + 80, 50);
      ctx.save();
      ctx.translate(452, 78);
      ctx.rotate(-0.025);
      ui.rr(ctx, -172, -38, 344, 76, 20);
      ctx.fillStyle = '#e8352e';
      ctx.fill();
      ctx.strokeStyle = '#ffffff';
      ctx.lineWidth = 4;
      ctx.stroke();
      ui.font(ctx, 46);
      ctx.fillStyle = '#ffffff';
      ctx.fillText('弹弹弹珠吧', 0, 2);
      ui.drawStarSticker(ctx, -146, 0, 9, 0, 0.95);
      ui.drawStarSticker(ctx, 146, 0, 9, 0, 0.95);
      ctx.restore();
    } else {
      var titleY = Math.max(196, (this.iconRowY || 42) + 64);   /* 跟随图标行,永不被按键遮挡 */
      ui.font(ctx, 46);
      ctx.fillStyle = C.panelText;
      ctx.fillText('弹弹弹珠吧', 160, titleY);
      ui.font(ctx, 17, false);
      ctx.fillStyle = C.dim;
      ctx.fillText('MARBLE MACHINE', 160, titleY + 40);
    }
```

```
/* 深色 LED 显示屏:弹珠库 / 倍率 */
this._ledBank = { x: 300, y: 200, w: 200, h: 84 };   /* 胶囊下方,躲开右上角 */
this._ledTier = { x: 512, y: 200, w: 140, h: 84 };
var leds = [this._ledBank, this._ledTier];
var labels = ['弹珠库', '倍率'];
for (var li = 0; li < 2; li++) {
  var lr = leds[li];
  ui.rr(ctx, lr.x, lr.y, lr.w, lr.h, 14);
  ctx.fillStyle = C.segBg;
  ctx.fill();
  ui.font(ctx, 16, false);
  ctx.fillStyle = C.ledLabel;
  ctx.fillText(labels[li], lr.x + lr.w / 2, lr.y + 20);
}
/* ---- 场地(浅暖灰斜面,铺满全宽;左右上角对称圆角与导轨重合) ---- */
var R2 = b.cornerL.r;
function fieldPath(c) {
  c.beginPath();
  c.moveTo(b.fieldX0 + R2, b.ceilingY);
  c.lineTo(W - 15 - R2, b.ceilingY);
  c.arcTo(W - 15, b.ceilingY, W - 15, b.ceilingY + R2, R2);
  c.lineTo(W - 15, b.floorY - 6);
  c.lineTo(b.fieldX0, b.floorY - 6);
  c.lineTo(b.fieldX0, b.ceilingY + R2);
  c.arcTo(b.fieldX0, b.ceilingY, b.fieldX0 + R2, b.ceilingY, R2);
  c.closePath();
}
fieldPath(ctx);
var field = ctx.createLinearGradient(0, b.ceilingY, 0, b.floorY);
field.addColorStop(0, C.fieldTop);
field.addColorStop(1, C.fieldBot);
ctx.fillStyle = field;
ctx.fill();
/* 场地印花与玻璃反光(复古主题) */
if (ARC) {
  ctx.save();
  fieldPath(ctx);
  ctx.clip();
  var deco = [[120, 470, 16], [380, 430, 13], [560, 500, 15], [210, 700, 18], [480, 760, 14
  ], [90, 950, 15], [340, 1000, 17], [560, 1010, 13], [160, 1150, 14], [420, 1180, 12]];
  for (var di = 0; di < deco.length; di++) {
    ui.drawStarSticker(ctx, deco[di][0], deco[di][1], deco[di][2], di * 0.7, 0.5);
  }
  ctx.globalAlpha = 0.28;
  ctx.fillStyle = C.glassShine;
  ctx.beginPath();
  ctx.moveTo(W * 0.16, b.ceilingY);
  ctx.lineTo(W * 0.4, b.ceilingY);
  ctx.lineTo(W * 0.12, b.floorY);
  ctx.lineTo(-W * 0.12, b.floorY);
```

```
    ctx.closePath();
    ctx.fill();
    ctx.beginPath();
    ctx.moveTo(W * 0.52, b.ceilingY);
    ctx.lineTo(W * 0.6, b.ceilingY);
    ctx.lineTo(W * 0.32, b.floorY);
    ctx.lineTo(W * 0.24, b.floorY);
    ctx.closePath();
    ctx.fill();
    ctx.globalAlpha = 1;
    ctx.restore();
  }
  /* ---- 发射道槽(场内右侧,自内墙顶以下;槽宽 = 内墙 face 到右墙 face) ---- */
  ctx.fillStyle = C.laneSlot;
  ctx.fillRect(b.lane.x0 + 8, b.lane.top, (W - 28) - (b.lane.x0 + 8), b.lane.floorY - b.lane.
    top);
  ctx.fillStyle = C.laneEdge;
  ctx.fillRect(b.lane.x0 - 7, b.lane.top, 14, b.lane.floorY - b.lane.top);
  /* 边界与导流一体轨:左墙 → 左弧角 → 顶轨 → 右弧角 → 右墙(竖直段与弧端同 x,严丝合缝) */
  ctx.save();
  ctx.lineCap = 'round';
  ctx.lineJoin = 'round';
  ctx.strokeStyle = C.trackMain;
  ctx.lineWidth = 14;
  ctx.beginPath();
  ctx.moveTo(b.fieldX0, b.floorY - 56);
  ctx.lineTo(b.fieldX0, b.cornerL.cy);
  ctx.arc(b.cornerL.cx, b.cornerL.cy, b.cornerL.r, Math.PI, Math.PI * 1.5, false);
  ctx.lineTo(b.cornerR.cx, b.ceilingY);
  ctx.arc(b.cornerR.cx, b.cornerR.cy, b.cornerR.r, Math.PI * 1.5, Math.PI * 2, false);
  ctx.lineTo(W - 21, b.floorY - 56);
  ctx.stroke();
  ctx.strokeStyle = C.trackHi;
  ctx.lineWidth = 3;
  ctx.stroke();
  ctx.restore();
  /* 墙边斜挡板(轨道同款材质,可见) */
  ctx.save();
  ctx.lineCap = 'round';
  ctx.strokeStyle = C.trackMain;
  ctx.lineWidth = 10;
  for (var wi = 0; wi < b.walls.length; wi++) {
    var wsg = b.walls[wi];
    if (wsg.kind !== 'deflector') continue;
    ctx.beginPath();
    ctx.moveTo(wsg.x1, wsg.y1);
    ctx.lineTo(wsg.x2, wsg.y2);
    ctx.stroke();
  }
  ctx.restore();
```

```
/* 弹道内墙顶圆帽 */
ctx.beginPath();
ctx.arc(b.lane.x0, b.lane.top, 8, 0, U.TAU);
ctx.fillStyle = C.capDot;
ctx.fill();
/* ---- 边框 LED 灯带(复古主题静态底) ---- */
if (ARC) {
  for (var bi = 0; bi < b.borderLeds.length; bi++) {
    var bd = b.borderLeds[bi];
    var col = bd.color === 'red' ? C.ledRed : C.ledGreen;
    ctx.beginPath();
    ctx.arc(bd.x, bd.y, 7, 0, U.TAU);
    ctx.fillStyle = C.ledDim;
    ctx.fill();
    ctx.beginPath();
    ctx.arc(bd.x, bd.y, 4.5, 0, U.TAU);
    ctx.fillStyle = col;
    ctx.globalAlpha = 0.55;
    ctx.fill();
    ctx.globalAlpha = 1;
  }
}
/* ---- 洞口条 ---- */
ui.rr(ctx, b.fieldX0 - 8, b.floorY - 58, b.fieldRight - b.fieldX0 + 16, 66, 16);
if (ARC) {
  var rail = ctx.createLinearGradient(0, b.floorY - 58, 0, b.floorY + 8);
  rail.addColorStop(0, C.railHi);
  rail.addColorStop(1, C.rail);
  ctx.fillStyle = rail;
  ctx.fill();
  ctx.strokeStyle = 'rgba(90,55,15,0.6)';
  ctx.lineWidth = 2;
  ctx.stroke();
} else {
  ctx.fillStyle = C.rail;
  ctx.fill();
}
for (var hi = 0; hi < b.holes.length; hi++) {
  var hk = b.holes[hi];
  ctx.beginPath();
  ctx.ellipse(hk.cx, b.floorY - 26, b.pw * 0.5 - 6, 14, 0, 0, U.TAU);
  ctx.fillStyle = C.holeDark;
  ctx.fill();
  ctx.strokeStyle = C.holeStroke;
  ctx.lineWidth = 1.5;
  ctx.stroke();
}
/* 隔针 */
for (var dvi = 0; dvi < b.dividers.length; dvi++) {
  var dv = b.dividers[dvi];
```

```
    ctx.strokeStyle = C.dividerLine;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(dv.x1, dv.y1);
    ctx.lineTo(dv.x1, b.floorY);
    ctx.stroke();
    ctx.beginPath();
    ctx.arc(dv.x1, dv.y1, 5.5, 0, U.TAU);
    ctx.fillStyle = C.capFill;
    ctx.fill();
    ctx.strokeStyle = C.capStroke;
    ctx.lineWidth = 1.5;
    ctx.stroke();
  }
  /* ---- 操作台:极简 = 浅色 + 发丝线;复古 = 黄渐变 + 白描边 ---- */
  var panelY = H - b.panelH;
  ui.rr(ctx, 8, panelY, W - 12, H - panelY - 4, 24);
  if (ARC) {
    var panel = ctx.createLinearGradient(0, panelY, 0, H);
    panel.addColorStop(0, C.cabYellowHi);
    panel.addColorStop(1, C.cabYellow);
    ctx.fillStyle = panel;
    ctx.fill();
    ctx.strokeStyle = C.frame;
    ctx.lineWidth = 6;
    ctx.stroke();
  } else {
    ctx.fillStyle = C.cabYellowHi;
    ctx.fill();
    ctx.strokeStyle = 'rgba(0,0,0,0.06)';
    ctx.lineWidth = 2;
    ctx.stroke();
  }
  /* 弹珠库托盘 */
  ctx.textAlign = 'left';
  ui.font(ctx, 22);
  ctx.fillStyle = C.panelText;
  ctx.fillText('弹珠库', 46, panelY + 40);
  /* GO 按钮底座 */
  this._go = { x: 195, y: panelY + 148, r: 44 };
  if (ARC) {
    ctx.beginPath();
    ctx.arc(this._go.x, this._go.y, this._go.r + 12, 0, U.TAU);
    var rim = ctx.createLinearGradient(this._go.x - 56, this._go.y - 56, this._go.x + 56, thi
    s._go.y + 56);
    rim.addColorStop(0, '#e8eaee');
    rim.addColorStop(0.5, '#b9bcc4');
    rim.addColorStop(1, '#8f939e');
    ctx.fillStyle = rim;
    ctx.fill();
```

```
      ctx.beginPath();
      ctx.arc(this._go.x, this._go.y, this._go.r, 0, U.TAU);
      ctx.fillStyle = 'rgba(70,40,8,0.3)';
      ctx.fill();
    } else {
      ctx.save();
      ctx.shadowColor = 'rgba(40,35,25,0.22)';
      ctx.shadowBlur = 18;
      ctx.shadowOffsetY = 6;
      ctx.beginPath();
      ctx.arc(this._go.x, this._go.y, this._go.r, 0, U.TAU);
      ctx.fillStyle = C.goBg;
      ctx.fill();
      ctx.restore();
    }
    /* 力度条 */
    this._bar = { x: 288, y: panelY + 56, w: 32, h: b.panelH - 118 };
    ui.rr(ctx, this._bar.x, this._bar.y, this._bar.w, this._bar.h, 10);
    ctx.fillStyle = C.barTrack;
    ctx.fill();
    if (ARC) {
      ctx.strokeStyle = C.panelText;
      ctx.lineWidth = 3;
      ctx.stroke();
    }
    ctx.textAlign = 'center';
    ui.font(ctx, 18, false);
    ctx.fillStyle = C.dim;
    ctx.fillText('力度', this._bar.x + this._bar.w / 2, this._bar.y - 16);
    /* 投入档位标签 */
    ctx.textAlign = 'center';
    ui.font(ctx, 20);
    ctx.fillStyle = C.panelText;
    ctx.fillText('投入弹珠(选倍率)', 536, panelY + 40);
    ui.font(ctx, 16, false);
    ctx.fillStyle = C.dim;
    ctx.fillText('投入越多 · 倍率越高 · 亮灯越少', 536, panelY + 148);
    /* 安全提示(操作台底栏小字) */
    ctx.textAlign = 'center';
    ui.font(ctx, 11, false);
    ctx.fillStyle = C.finePrint;
    ctx.fillText('本游戏适合8周岁以上用户使用 · 弹珠只供游戏，严禁吞食', W / 2, panelY + 202);
    /* 提示 */
    ui.font(ctx, 20, false);
    ctx.fillStyle = ARC ? C.panelText : C.dim;
    ctx.fillText('按住蓄力 · 松手弹珠', this._go.x, panelY + b.panelH - 20);
    this.bg = off;
};
/* ================= 渲染 ================= */
Game.prototype.render = function () {
```

```
  var ctx = this.ctx;
  var p = this.platform;
  ctx.setTransform(1, 0, 0, 1, 0, 0);
  if (this.bg) {
    ctx.drawImage(this.bg, 0, 0);
  } else {
    ctx.fillStyle = C.cabYellow;
    ctx.fillRect(0, 0, p.canvas.width, p.canvas.height);
  }
  var s = this.view.scale * p.dpr;
  ctx.setTransform(s, 0, 0, s,
    (this.view.offX + this.fx.sx * this.view.scale) * p.dpr,
    (this.view.offY + this.fx.sy * this.view.scale) * p.dpr);
  this.frameButtons.length = 0;
  var st = this.state;   /* 先取状态再画图标(此前 var 提升导致判断时 st 为 undefined) */
  /* 顶部图标簇:音效 / 音乐 / 主页(主页非菜单态显示) */
  MG.ui.iconButton(this, 56, this.iconRowY, 24, 'sound', function () { this.toggleMute(); }.b
    ind(this));
  MG.ui.iconButton(this, 106, this.iconRowY, 24, 'music', function () { this.cycleMusic(); }.
    bind(this));
  /* 抖音:侧边栏复访入口(审核必接能力):菜单态在主页后,游戏态在暂停后 */
  if (p.name === 'douyin' && st === 'menu') {
    MG.ui.iconButton(this, 206, this.iconRowY, 24, 'sidebar', function () { this.platform.add
    ToSidebar(); }.bind(this));
  }
  if (st !== 'menu') {
    MG.ui.iconButton(this, 156, this.iconRowY, 24, 'home', function () { this.backToMenu(); }
    .bind(this));
  }
  if (st === 'menu') {
    MG.ui.renderMenu(this, false);
  } else if (st === 'help') {
    MG.ui.renderMenu(this, true);
    MG.ui.renderHelp(this);
  } else {
    this.renderPlay();
    if (st === 'pause') MG.ui.renderPause(this);
    else if (st === 'result') MG.ui.renderResult(this);
  }
};
Game.prototype.renderPlay = function () {
  var ctx = this.ctx;
  var p = this.platform;   /* 侧边栏图标判断用(此前未声明,进入游戏即抛 TypeError 冻死画面) */
  var b = this.board;
  var W = this.view.W, H = this.view.H;
  var t = this.time;
  var ui = MG.ui;
  var i;
  /* ---- 背板:LED 数值 + 小按钮 ---- */
  this.drawSeg(this._ledBank, String(this.bank));
```

```
this.drawSeg(this._ledTier, '×' + this.tier().mult);
ui.iconButton(this, 206, this.iconRowY, 24, 'pause', function () { this.state = 'pause'; th
  is.stateT = 0; MG.music.pause(); }.bind(this));
if (p.name === 'douyin') {
  ui.iconButton(this, 256, this.iconRowY, 24, 'sidebar', function () { this.platform.addToS
  idebar(); }.bind(this));
}
/* ---- 弹道箭头 ---- */
var gap = 90;
var anim = (t * 170) % gap;
ctx.save();
  ctx.fillStyle = C.laneArrow;
for (i = 0; i < 5; i++) {
  var ay = b.lane.floorY - 60 - i * gap + anim;
  if (ay < b.lane.top + 30 || ay > b.lane.floorY - 20) continue;
  ctx.beginPath();
  ctx.moveTo(b.lane.cx - 9, ay + 11);
  ctx.lineTo(b.lane.cx, ay - 9);
  ctx.lineTo(b.lane.cx + 9, ay + 11);
  ctx.closePath();
  ctx.fill();
}
ctx.restore();
/* ---- 边框 LED 追逐灯(复古主题) ---- */
if (CFG.themeName === 'arcade') {
  var bl = b.borderLeds;
  var head = Math.floor(t * 7) % bl.length;
  for (var k = 0; k < 3; k++) {
    var idx = (head + k) % bl.length;
    var bd = bl[idx];
    var col = bd.color === 'red' ? C.ledRed : C.ledGreen;
    ctx.save();
    ctx.shadowColor = col;
    ctx.shadowBlur = 12;
    ctx.beginPath();
    ctx.arc(bd.x, bd.y, 5.5, 0, U.TAU);
    ctx.fillStyle = '#ffffff';
    ctx.globalAlpha = 0.9 - k * 0.25;
    ctx.fill();
    ctx.restore();
  }
}
/* ---- 拉杆闪亮(发射瞬间) ---- */
if (this.launcherPulse > 0) {
  ctx.save();
  ctx.globalAlpha = this.launcherPulse * 0.55;
  ctx.shadowColor = C.pulseColor;
  ctx.shadowBlur = 16;
  ctx.beginPath();
  ctx.arc(b.lane.cx, b.lane.floorY - 30, 17, 0, U.TAU);
```

```
    ctx.fillStyle = C.pulseColor;
    ctx.fill();
    ctx.restore();
  }
  /* ---- 钉子:复古 = 部分红绿 LED 钉;极简 = 白圆 + 发丝圈 ---- */
  for (i = 0; i < b.pegs.length; i++) {
    var p = b.pegs[i];
    if (CFG.themeName === 'arcade' && p.led) {
      var lc = p.led === 'red' ? C.ledRed : C.ledGreen;
      var pulse = 0.5 + 0.5 * Math.sin(t * 2.2 + i * 1.7);
      ctx.save();
      ctx.shadowColor = lc;
      ctx.shadowBlur = 8 + pulse * 6;
      ctx.beginPath();
      ctx.arc(p.x, p.y, p.r, 0, U.TAU);
      ctx.fillStyle = lc;
      ctx.fill();
      ctx.restore();
      ctx.beginPath();
      ctx.arc(p.x - p.r * 0.3, p.y - p.r * 0.3, p.r * 0.3, 0, U.TAU);
      ctx.fillStyle = 'rgba(255,255,255,0.85)';
      ctx.fill();
    } else {
      ctx.beginPath();
      ctx.arc(p.x + 1, p.y + 2, p.r, 0, U.TAU);
      ctx.fillStyle = 'rgba(0,0,0,0.07)';
      ctx.fill();
      ctx.beginPath();
      ctx.arc(p.x, p.y, p.r, 0, U.TAU);
      ctx.fillStyle = C.pegBody;
      ctx.fill();
      ctx.strokeStyle = C.pegRim;
      ctx.lineWidth = 1.5;
      ctx.stroke();
    }
    if (p.flash > 0) {
      ctx.globalAlpha = p.flash * 0.9;
      ctx.fillStyle = C.pegFlash;
      ctx.beginPath();
      ctx.arc(p.x, p.y, p.r + 3, 0, U.TAU);
      ctx.fill();
      ctx.globalAlpha = 1;
    }
  }
  /* ---- 大弹柱:复古 = 玻璃渐变球;极简 = 白圆环 ---- */
  for (i = 0; i < b.bumpers.length; i++) {
    var bp = b.bumpers[i];
    var r = bp.r + bp.pulse * 5;
    if (CFG.themeName === 'arcade') {
      var bg2 = ctx.createRadialGradient(bp.x - r * 0.3, bp.y - r * 0.3, 2, bp.x, bp.y, r);
```

```
      bg2.addColorStop(0, 'rgba(255,255,255,0.95)');
      bg2.addColorStop(0.6, 'rgba(235,225,200,0.9)');
      bg2.addColorStop(1, 'rgba(180,150,95,0.9)');
      ctx.beginPath();
      ctx.arc(bp.x, bp.y, r, 0, U.TAU);
      ctx.fillStyle = bg2;
      ctx.fill();
      ctx.strokeStyle = C.bumperRing;
      ctx.lineWidth = 3;
      ctx.stroke();
    } else {
      ctx.beginPath();
      ctx.arc(bp.x + 1.5, bp.y + 2.5, r, 0, U.TAU);
      ctx.fillStyle = 'rgba(0,0,0,0.08)';
      ctx.fill();
      ctx.beginPath();
      ctx.arc(bp.x, bp.y, r, 0, U.TAU);
      ctx.fillStyle = '#ffffff';
      ctx.fill();
      ctx.strokeStyle = C.bumperRing;
      ctx.lineWidth = 2.5;
      ctx.stroke();
      ctx.beginPath();
      ctx.arc(bp.x, bp.y, r * 0.55, 0, U.TAU);
      ctx.strokeStyle = 'rgba(0,0,0,0.08)';
      ctx.lineWidth = 2;
      ctx.stroke();
    }
  }
  /* ---- 亮灯洞(动态辉光) ---- */
  for (i = 0; i < b.holes.length; i++) {
    var hk = b.holes[i];
    if (hk.lit) {
      var pulse2 = 0.5 + 0.5 * Math.sin(t * 3.4 + i);
      ctx.save();
      ctx.shadowColor = C.holeLit;
      ctx.shadowBlur = 16 + pulse2 * 10;
      ctx.beginPath();
      ctx.ellipse(hk.cx, b.floorY - 26, b.pw * 0.5 - 6, 14, 0, 0, U.TAU);
      ctx.fillStyle = C.holeLit;
      ctx.fill();
      ctx.restore();
      ctx.beginPath();
      ctx.ellipse(hk.cx, b.floorY - 26, (b.pw * 0.5 - 6) * 0.45, 5.5, 0, 0, U.TAU);
      ctx.fillStyle = '#ffd9c9';
      ctx.fill();
    }
    if (hk.flash > 0) {
      ctx.globalAlpha = hk.flash * 0.85;
      ctx.fillStyle = '#ffffff';
```

```
      ctx.beginPath();
      ctx.ellipse(hk.cx, b.floorY - 26, b.pw * 0.5 - 6, 14, 0, 0, U.TAU);
      ctx.fill();
      ctx.globalAlpha = 1;
    }
  }
  /* ---- 玻璃弹珠 ---- */
  var tints = C.marbleTints;
  var balls = this.world.balls;
  for (i = 0; i < balls.length; i++) {
    var m = balls[i];
    if (m.dead) continue;
    var tintC = tints[m.skin % tints.length];
    for (var k2 = 0; k2 < m.trail.length; k2++) {
      var tr = m.trail[k2];
      var a = (k2 + 1) / m.trail.length;
      ctx.globalAlpha = a * 0.16;
      ctx.fillStyle = tintC;
      ctx.beginPath();
      ctx.arc(tr.x, tr.y, m.r * (0.3 + 0.5 * a), 0, U.TAU);
      ctx.fill();
    }
    ctx.globalAlpha = 1;
    ui.drawMarble(ctx, m.x, m.y, m.r, tintC);
  }
  /* ---- 特效(裁剪进机台:粒子不许飞到机台外的留黑区) ---- */
  ctx.save();
  ui.rr(ctx, 2, 2, W - 4, H - 4, 34);
  ctx.clip();
  MG.fx.render(ctx, this.fx);
  ctx.restore();
  /* ---- 操作台动态:档位按钮 / GO / 力度条 / 托盘 ---- */
  /* 投入档位按钮(极简:白卡 / 墨色选中) */
  var cw = 70, ch = 64, gap2 = 5;
  var cx0 = 536 - (CFG.tiers.length * (cw + gap2) - gap2) / 2;
  for (i = 0; i < CFG.tiers.length; i++) {
    var tp = CFG.tiers[i];
    var cx = cx0 + i * (cw + gap2);
    var cy = H - b.panelH + 62;
    var selected = i === this.selIdx;
    var disabled = tp.cost > this.bank;
    ui.rr(ctx, cx, cy, cw, ch, 14);
    if (selected) {
      ctx.fillStyle = C.selBg;
      ctx.fill();
    } else if (disabled) {
      ctx.fillStyle = C.pillDis;
      ctx.fill();
    } else {
      ctx.fillStyle = C.pillBg;
```

```
      ctx.fill();
    }
    ctx.strokeStyle = selected ? C.selBg : C.pillLine;
    ctx.lineWidth = 1.5;
    ctx.stroke();
    ctx.textAlign = 'center';
    ctx.textBaseline = 'middle';
    ui.font(ctx, 17, false);
    ctx.fillStyle = selected ? C.selSub : (disabled ? C.pillDisText : C.dim);
    ctx.fillText(tp.cost + '颗', cx + cw / 2, cy + 18);
    ui.font(ctx, 26);
    ctx.fillStyle = selected ? '#ffffff' : (disabled ? C.pillDisText : C.pillText);
    ctx.fillText('×' + tp.mult, cx + cw / 2, cy + 44);
    if (!disabled) {
      (function (idx) {
        this.frameButtons.push({ x: cx, y: cy, w: cw, h: ch, cb: function () { this.selectTie
    r(idx); }.bind(this) });
      }).call(this, i);
    }
  }
  var power = this.charging ? this.chargePower() : 0;
  var pc = power < 0.4 ? C.pow1 : (power < 0.8 ? C.pow2 : C.pow3);
  /* 力度条填充 */
  if (power > 0.01) {
    var bar = this._bar;
    var fh = (bar.h - 8) * power;
    ui.rr(ctx, bar.x + 4, bar.y + bar.h - 4 - fh, bar.w - 8, fh, 7);
    ctx.fillStyle = pc;
    ctx.fill();
  }
  /* GO 按钮:复古 = 红渐变 + 白圈;极简 = 墨色圆 */
  var go = this._go;
  var pressed = this.charging;
  ctx.beginPath();
  ctx.arc(go.x, go.y, pressed ? go.r - 3 : go.r, 0, U.TAU);
  if (CFG.themeName === 'arcade') {
    var gg = ctx.createRadialGradient(go.x - 12, go.y - 14, 6, go.x, go.y, go.r);
    gg.addColorStop(0, pressed ? '#ff8f7a' : '#ff6a52');
    gg.addColorStop(1, pressed ? C.goPress : C.goBg);
    ctx.fillStyle = gg;
    ctx.fill();
    ctx.strokeStyle = C.goRing;
    ctx.lineWidth = 3;
    ctx.stroke();
  } else {
    ctx.fillStyle = pressed ? C.goPress : C.goBg;
    ctx.fill();
  }
  ui.font(ctx, 30);
  ctx.textAlign = 'center';
```

```
    ctx.textBaseline = 'middle';
    ctx.fillStyle = '#ffffff';
    ctx.fillText('GO', go.x, go.y + 1);
    if (this.charging) {
      ctx.beginPath();
      ctx.arc(go.x, go.y, go.r + 10, -Math.PI / 2, -Math.PI / 2 + power * U.TAU * 0.83);
      ctx.strokeStyle = pc;
      ctx.lineWidth = 6;
      ctx.lineCap = 'round';
      ctx.stroke();
      ctx.lineCap = 'butt';
    }
    if (this.launcherPulse > 0) {
      ctx.beginPath();
      ctx.arc(go.x, go.y, go.r + 10 + this.launcherPulse * 14, 0, U.TAU);
      ctx.strokeStyle = C.pulseRing + (0.45 * this.launcherPulse) + ')';
      ctx.lineWidth = 4;
      ctx.stroke();
    }
    /* 弹珠库托盘点位 */
    var n2 = Math.min(this.bank, 6);
    for (i = 0; i < n2; i++) {
      var tx = 50 + (i % 3) * 34;
      var ty = H - b.panelH + 86 + Math.floor(i / 3) * 36;
      ui.drawMarble(ctx, tx, ty, 12, tints[i % tints.length]);
    }
    if (this.bank > 6) {
      ui.font(ctx, 22);
      ctx.textAlign = 'left';
      ctx.fillStyle = C.panelText;
      ctx.fillText('×' + this.bank, 50, H - b.panelH + 168);
    }
  };
  /* 红色 LED 数码管 */
  Game.prototype.drawSeg = function (rect, text) {
    var ctx = this.ctx;
    ctx.save();
    ctx.textAlign = 'center';
    ctx.textBaseline = 'middle';
    ctx.translate(rect.x + rect.w / 2, rect.y + rect.h * 0.62);
    MG.ui.segFont(ctx, rect.w > 180 ? 52 : 46);
    ctx.shadowColor = C.segRed;
    ctx.shadowBlur = 16;
    ctx.fillStyle = C.segRed;
    ctx.fillText(text, 0, 0);
    ctx.restore();
  };
  /* ================= 主循环 ================= */
  Game.prototype.frame = function (dt) {
    this.update(dt);
```

```
    this.render();
  };
  Game.prototype.start = function () {
    var self = this;
    this._last = 0;
    /* RAF 三级兜底:全局 → canvas.requestAnimationFrame → 定时器 */
    var g = typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? w
      indow : null);
    var cvs = this.platform.canvas;
    var raf = null;
    if (typeof requestAnimationFrame === 'function') raf = requestAnimationFrame;
    else if (cvs && typeof cvs.requestAnimationFrame === 'function') raf = function (cb) { cvs.
      requestAnimationFrame(cb); };
    else if (g && typeof g.requestAnimationFrame === 'function') raf = function (cb) { g.reques
      tAnimationFrame(cb); };
    function loop(ts) {
      /* 先排下一帧再画本帧:任何一帧 update/render 抛异常,
         都只跳过该帧并记录,绝不永久冻死主循环 */
      if (raf) raf(loop);
      var t = (ts || Date.now()) / 1000;
      if (!self._last) self._last = t;
      var dt = Math.min(0.05, t - self._last);
      self._last = t;
      try {
        self.frame(dt);
      } catch (e) {
        var msg = String((e && e.stack) || e);
        if (!self.errors) self.errors = [];
        if (self.errors.indexOf(msg) < 0 && self.errors.length < 20) self.errors.push(msg);
        try { console.error('[MG frame]', e); } catch (e2) { }
      }
    }
    if (raf) raf(loop);
    else setInterval(function () { loop(Date.now()); }, 16);
  };
  MG.Game = Game;
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
/* ================================================== */
/* 文件: ./src/main.js */
/* 弹弹珠机 - 启动入口(三端共用) */
(function (root) {
  'use strict';
  var MG = root.MG || (root.MG = {});
  function boot() {
    var platform = MG.platform.create();
    platform.setPreferredFramesPerSecond(60);
    var game = new MG.Game(platform);
    /* 调试句柄:小游戏环境无 window,用 GameGlobal */
    try {
```

```
      if (typeof window !== 'undefined') window.__mgGame = game;
      else if (typeof GameGlobal !== 'undefined') GameGlobal.__mgGame = game;
    } catch (e) { }
    game.start();
    MG.__game = game; /* 调试用 */
    return game;
  }
  MG.main = { boot: boot };
  if (typeof document !== 'undefined') {
    if (document.readyState === 'loading') {
      document.addEventListener('DOMContentLoaded', boot);
    } else {
      boot();
    }
  } else {
    boot(); /* 小游戏环境无 document,直接启动 */
  }
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
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  }
})(typeof GameGlobal !== 'undefined' ? GameGlobal : (typeof window !== 'undefined' ? window : (
      typeof globalThis !== 'undefined' ? globalThis : this)));
```